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Yanagisawa

(10) **Patent No.: US 12,411,399 B2**
(45) **Date of Patent: Sep. 9, 2025**

(54) **PROJECTION SYSTEM AND PROJECTOR**

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(73) Assignee: **SEIKO EPSON CORPORATION**, Tokyo (JP)

(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 426 days.

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(22) Filed: **Jan. 19, 2023**

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(30) **Foreign Application Priority Data**

Jan. 19, 2022 (JP) 2022-006175

(51) **Int. Cl.**
G03B 21/28 (2006.01)
G02B 13/16 (2006.01)

(52) **U.S. Cl.**
CPC ***G03B 21/28*** (2013.01); ***G02B 13/16*** (2013.01)

(58) **Field of Classification Search**
CPC G02B 13/18; G02B 13/16; G02B 17/08; G02B 17/0844; G02B 17/0812; G02B 17/0852; G02B 17/0848; G02B 17/0828; G03B 21/10; G03B 21/28
See application file for complete search history.

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(74) *Attorney, Agent, or Firm* — Oliff PLC

(57) **ABSTRACT**

A projection system includes a first and second optical system arranged from a reduction side toward an enlargement side. The second optical system includes an optical element having a concave reflection surface and a first lens having negative power, the optical element and first lens arranged from reduction side toward enlargement side. The projection system satisfies the following expressions:

$$TR \leq 0.3 \quad (1)$$

$$35 \leq (OAL/imy) \times (LL/imy) \times TR \times (1/NA) \leq 60 \quad (2)$$

OAL represents an axial inter-surface spacing from an image formation device to the reflection surface, imy represents a first distance from an optical axis to the largest image height at the image formation device, LL represents the largest radius of the first lens, TR represents a throw ratio, and NA represents the numerical aperture of the image formation device.

7 Claims, 21 Drawing Sheets

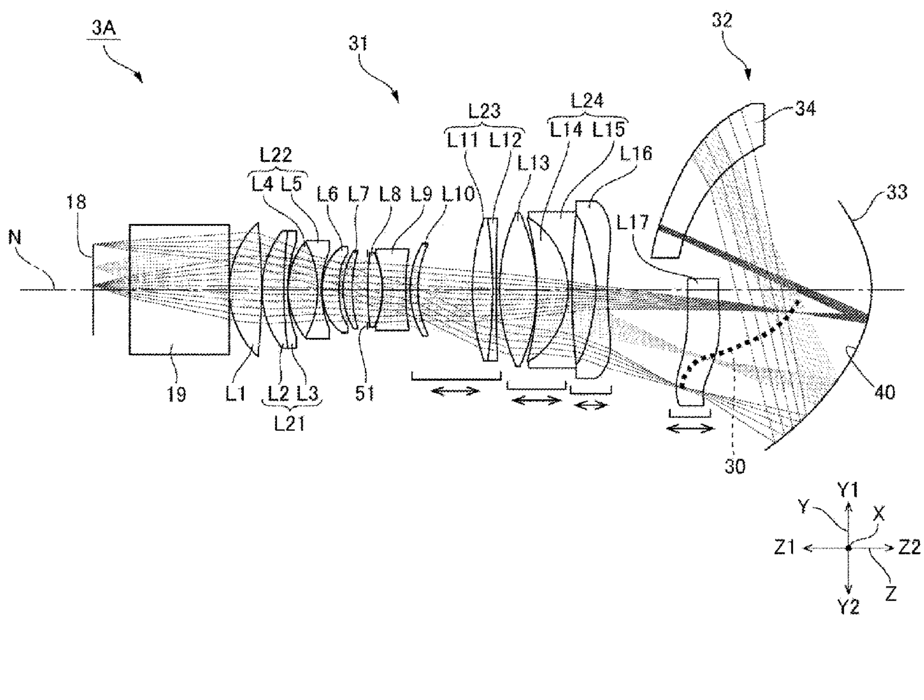
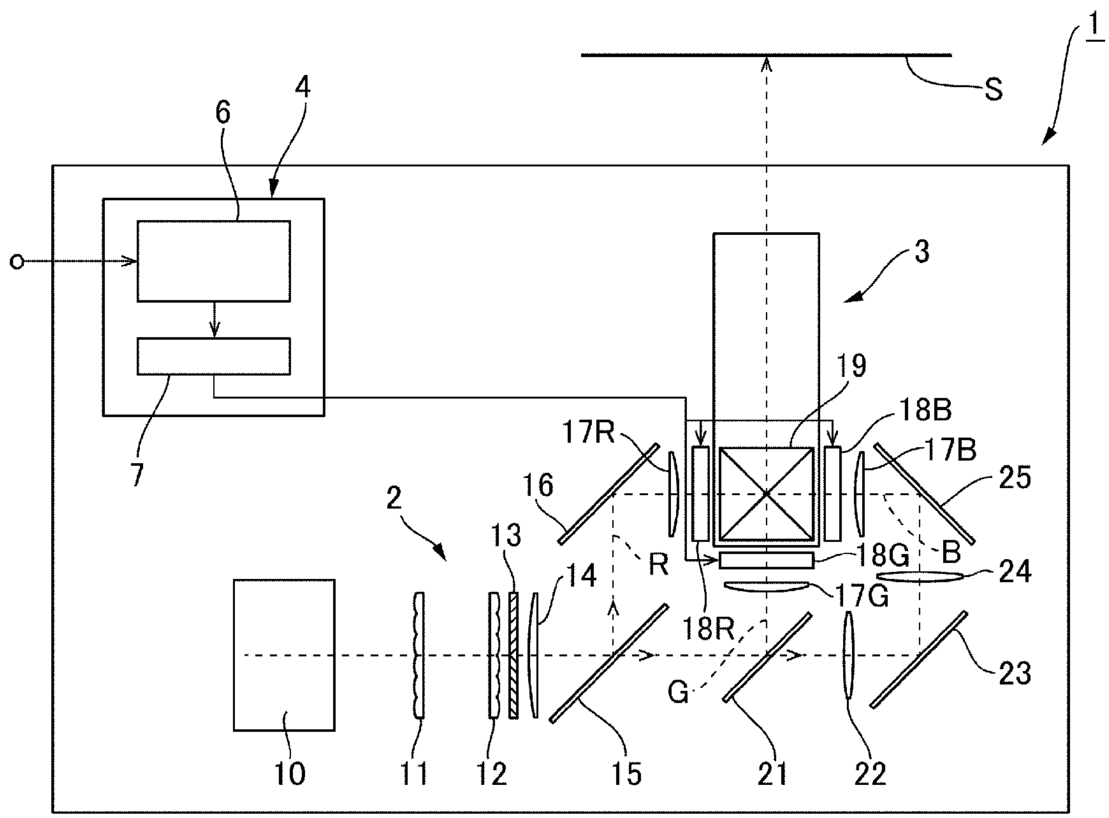


FIG. 1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17R
17B
16
25
2
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18G
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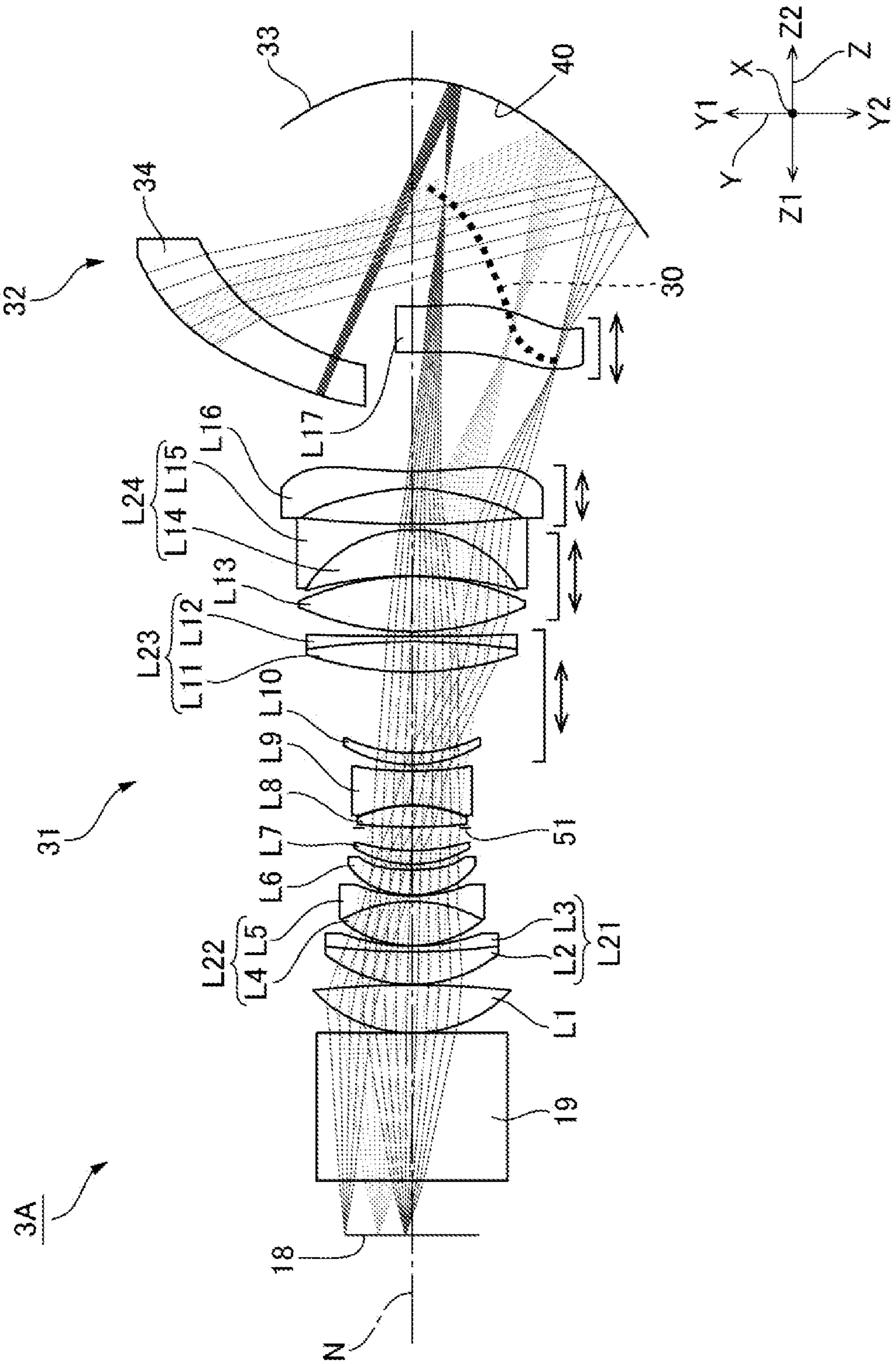
17G
18R
G
23
10
11
12
15
21
22
7

FIG. 2



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L17
L16
L24
L15
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51
L7
L6
L22
L5
L4
L3
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3A
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N
X
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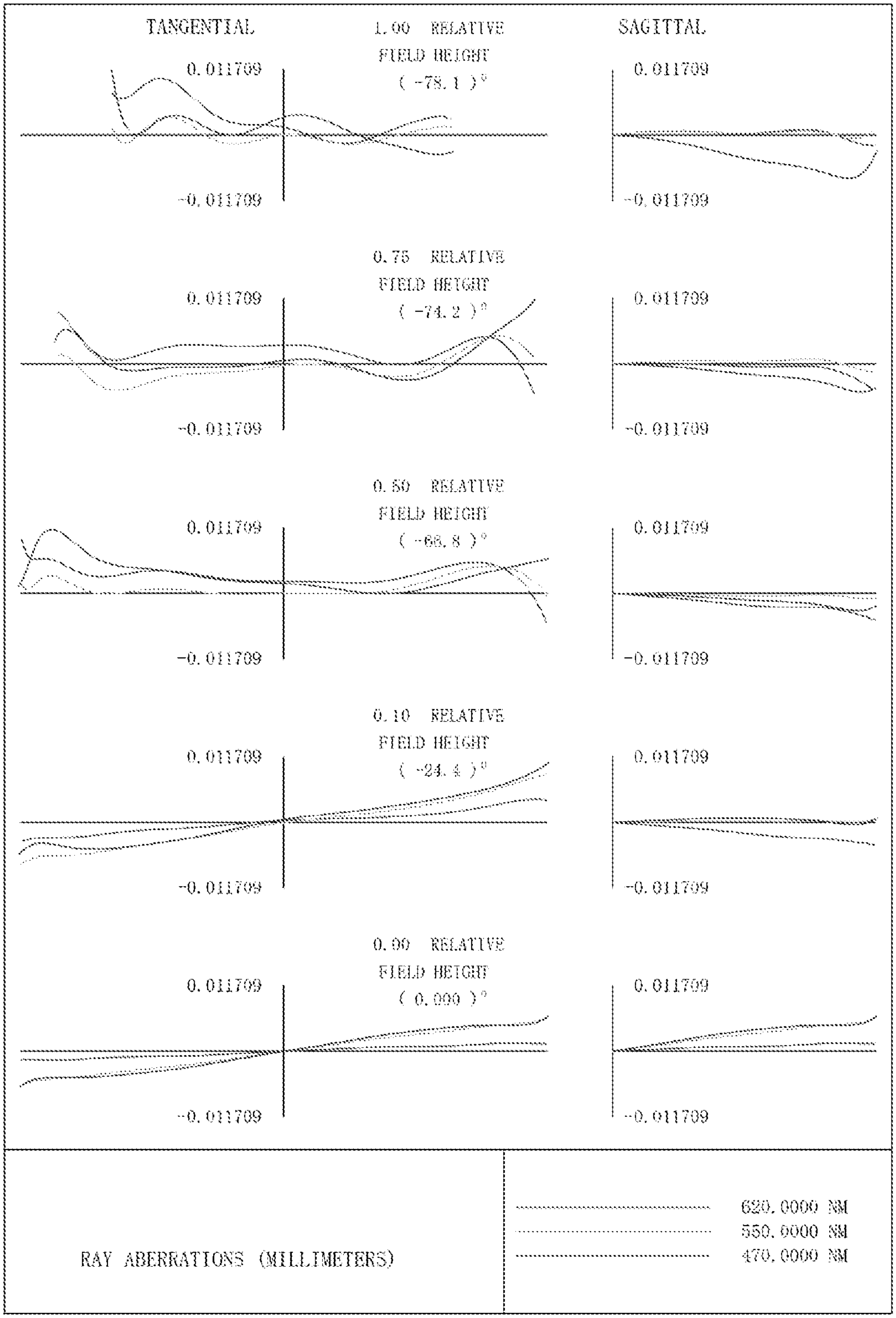
Z
Y1
Y2
Z1
Z2

FIG. 3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FIELD HEIGHT
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0.50 RELATIVE
FIELD HEIGHT
(-66.8)°
0.10 RELATIVE
FIELD HEIGHT
(-24.4)°
0.00 RELATIVE
FIELD HEIGHT
(0.000)°
RAY ABERRATIONS (MILLIMETERS)
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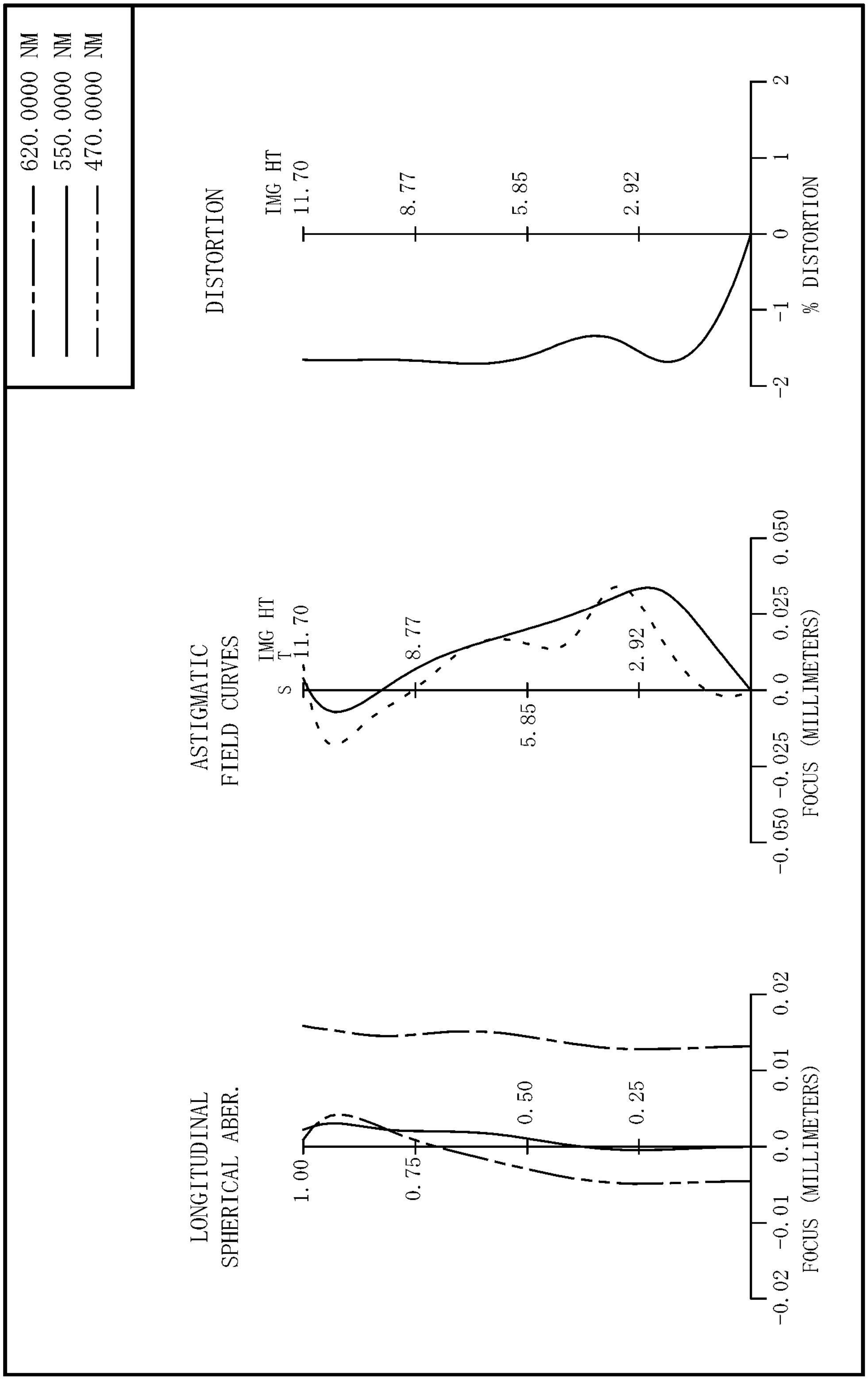
550.0000 NM
470.0000 NM

FIG. 4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0.25
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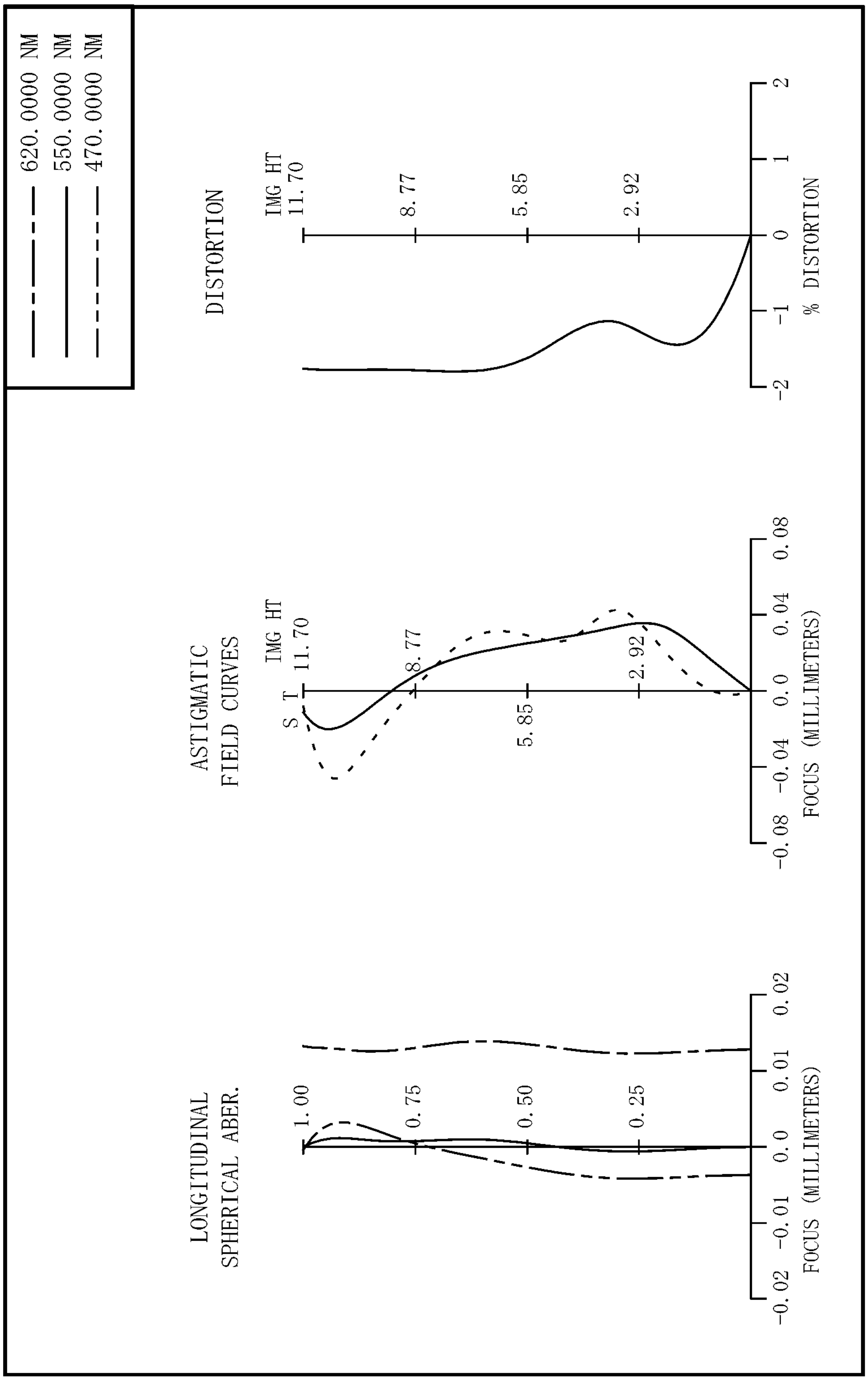
FOCUS (MILLIMETERS)
ASTIGMATIC FIELD CURVES
IMG HT
S
T
11.70
8.77
5.85
2.92
−0.050 −0.025 0.0 0.025 0.050
FOCUS (MILLIMETERS)
DISTORTION
IMG HT
11.70
8.77
5.85
2.92
−2 −1 0 1 2
% DISTORTION

FIG. 5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0.25
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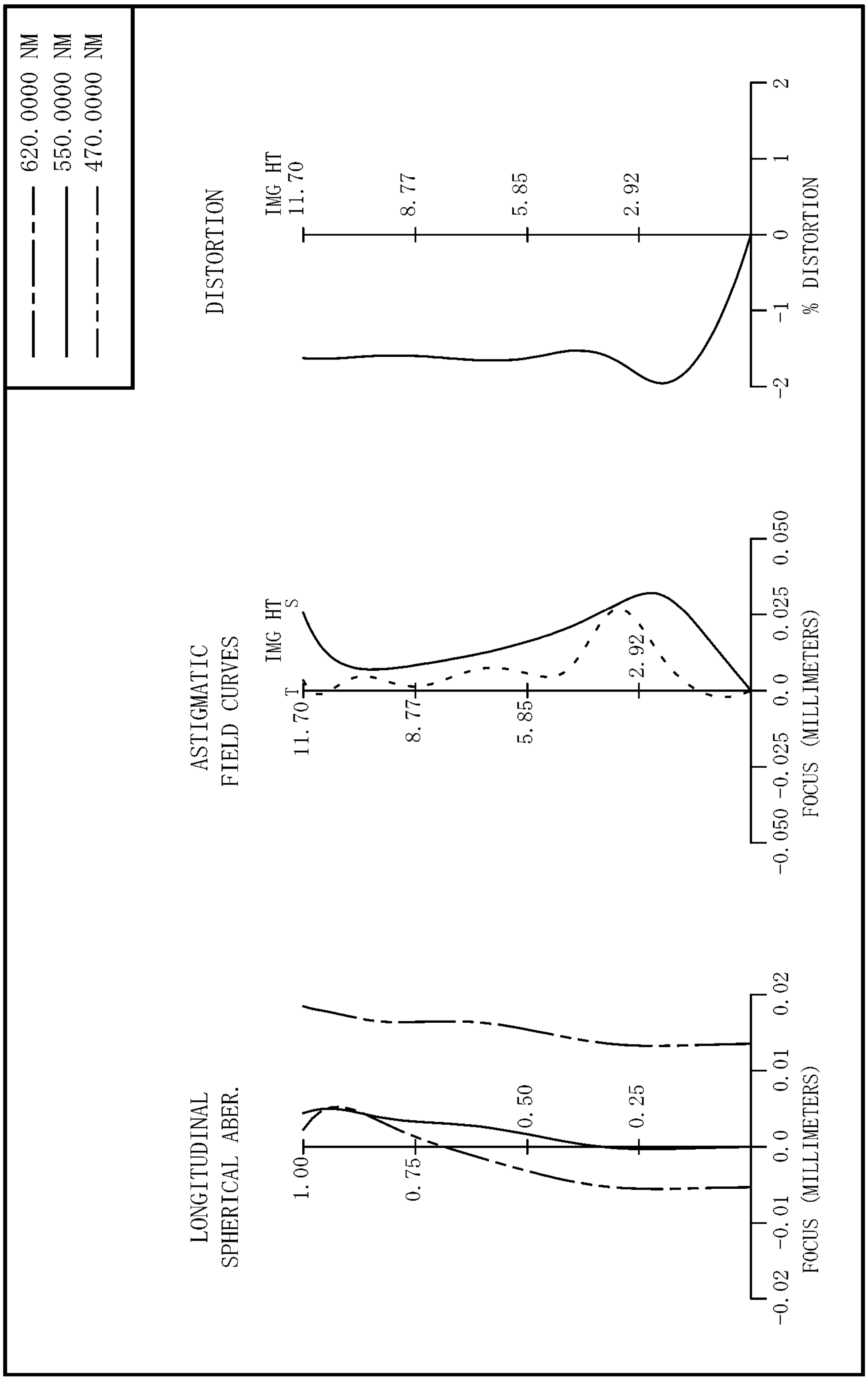
FOCUS (MILLIMETERS)
ASTIGMATIC FIELD CURVES
IMG HT
11.70
S T
8.77
5.85
2.92
−0.08 −0.04 0.0 0.04 0.08
FOCUS (MILLIMETERS)
DISTORTION
IMG HT
11.70
8.77
5.85
2.92
−2 −1 0 1 2
% DISTORTION

FIG. 6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0.50
0.25
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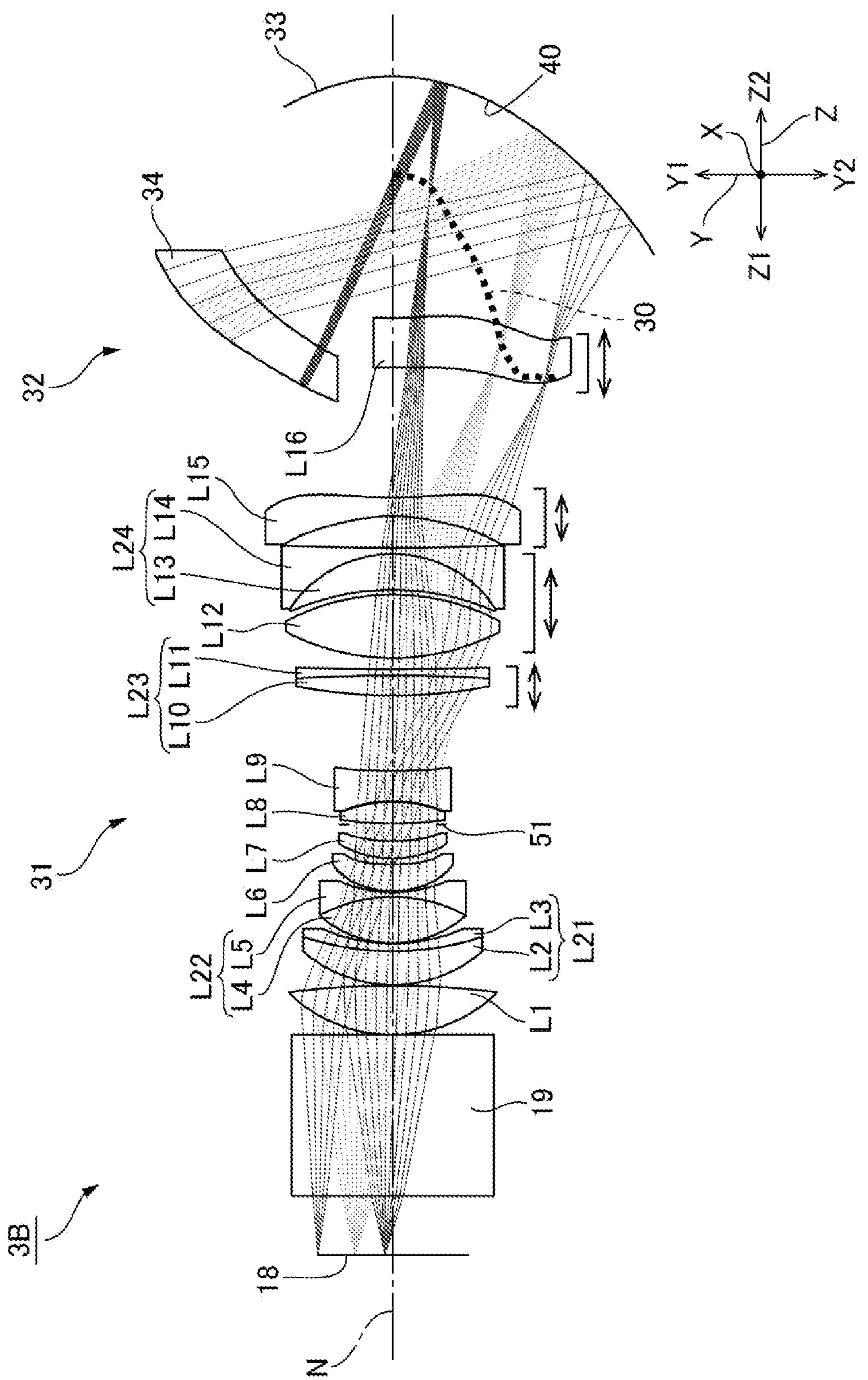
ASTIGMATIC FIELD CURVES
IMG HT
S
T
11.70
8.77
5.85
2.92
-0.050 -0.025 0.0 0.025 0.050
FOCUS (MILLIMETERS)
DISTORTION
IMG HT
11.70
8.77
5.85
2.92
-2 -1 0 1 2
% DISTORTION

FIG. 7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Z2
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Y1
Y2
Y
Z1
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32
L16
L15
L24
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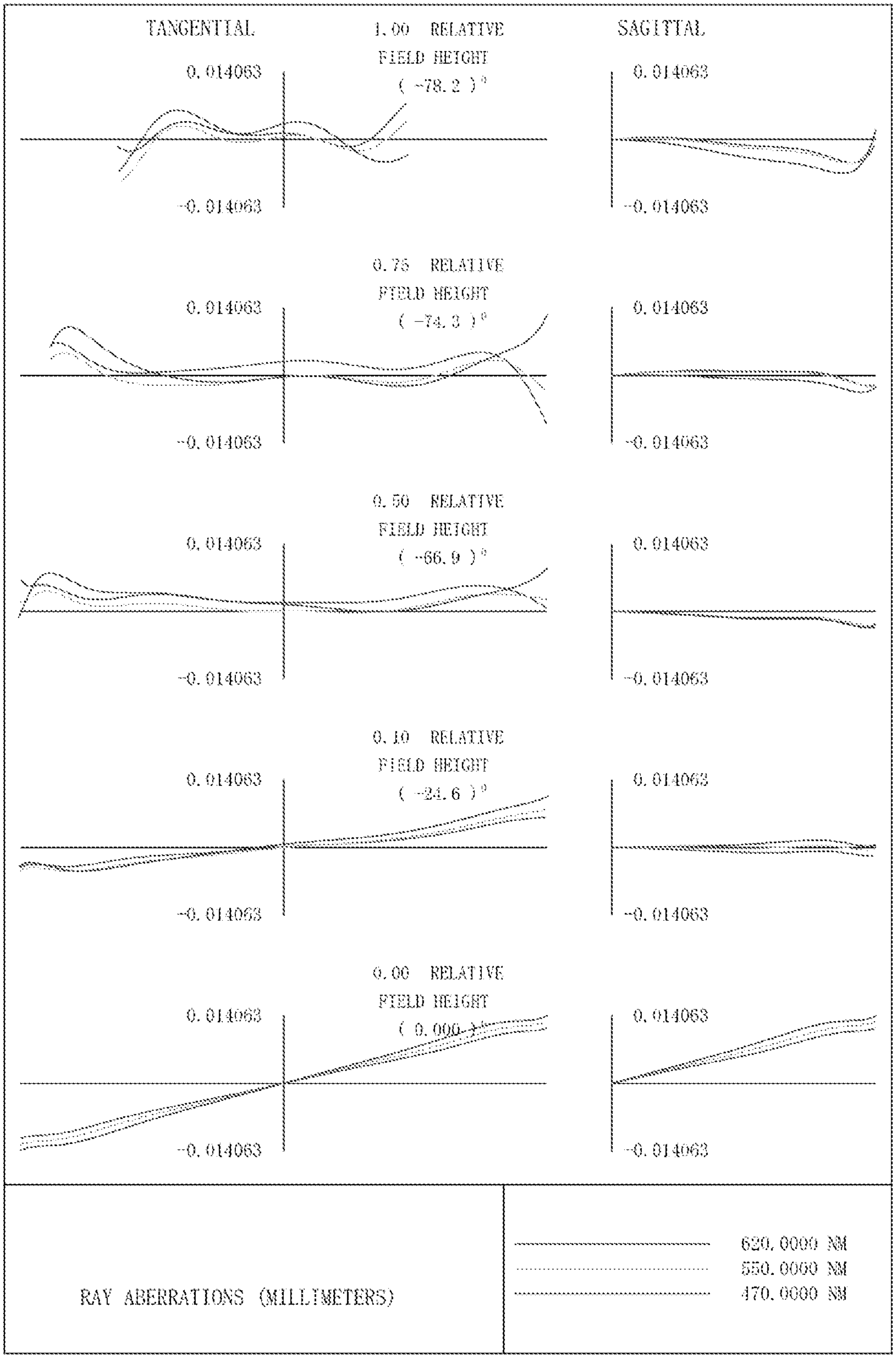
L13
L12
L23
L11
L10
L9
L8
L7
L6
L5
L4
L22
51
31
L3
L2
L21
L1
19
3B
18
N

FIG. 8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FIELD HEIGHT
(-74.3)°
0.50 RELATIVE
FIELD HEIGHT
(-66.9)°
0.10 RELATIVE
FIELD HEIGHT
(-24.6)°
0.00 RELATIVE
FIELD HEIGHT
(0.000)°
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-0.014063
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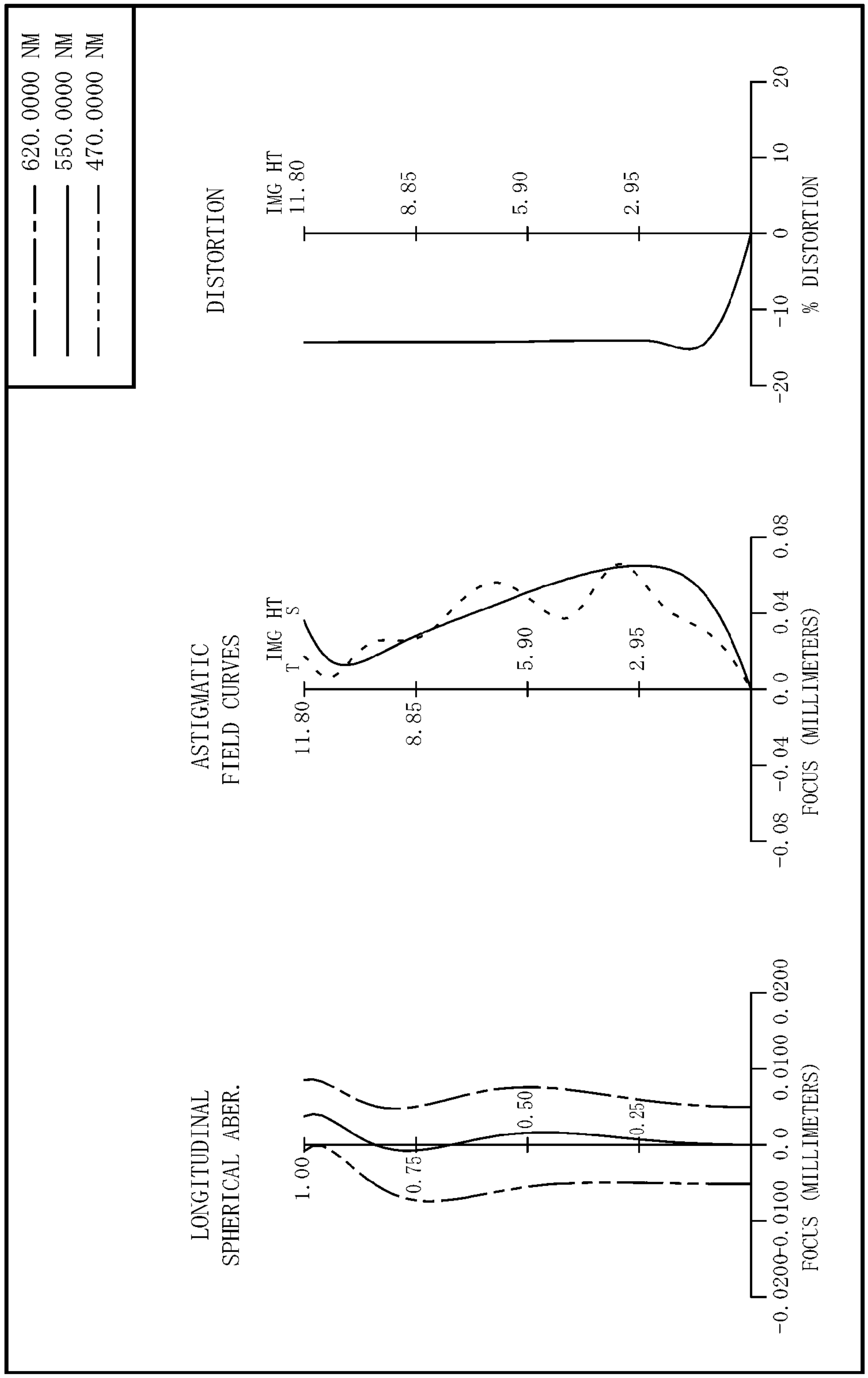
FIG. 9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0.0100
0.0200
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S
T
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8.85
5.90
2.95
-0.08
-0.04
0.0
0.04
0.08
FOCUS (MILLIMETERS)
DISTORTION
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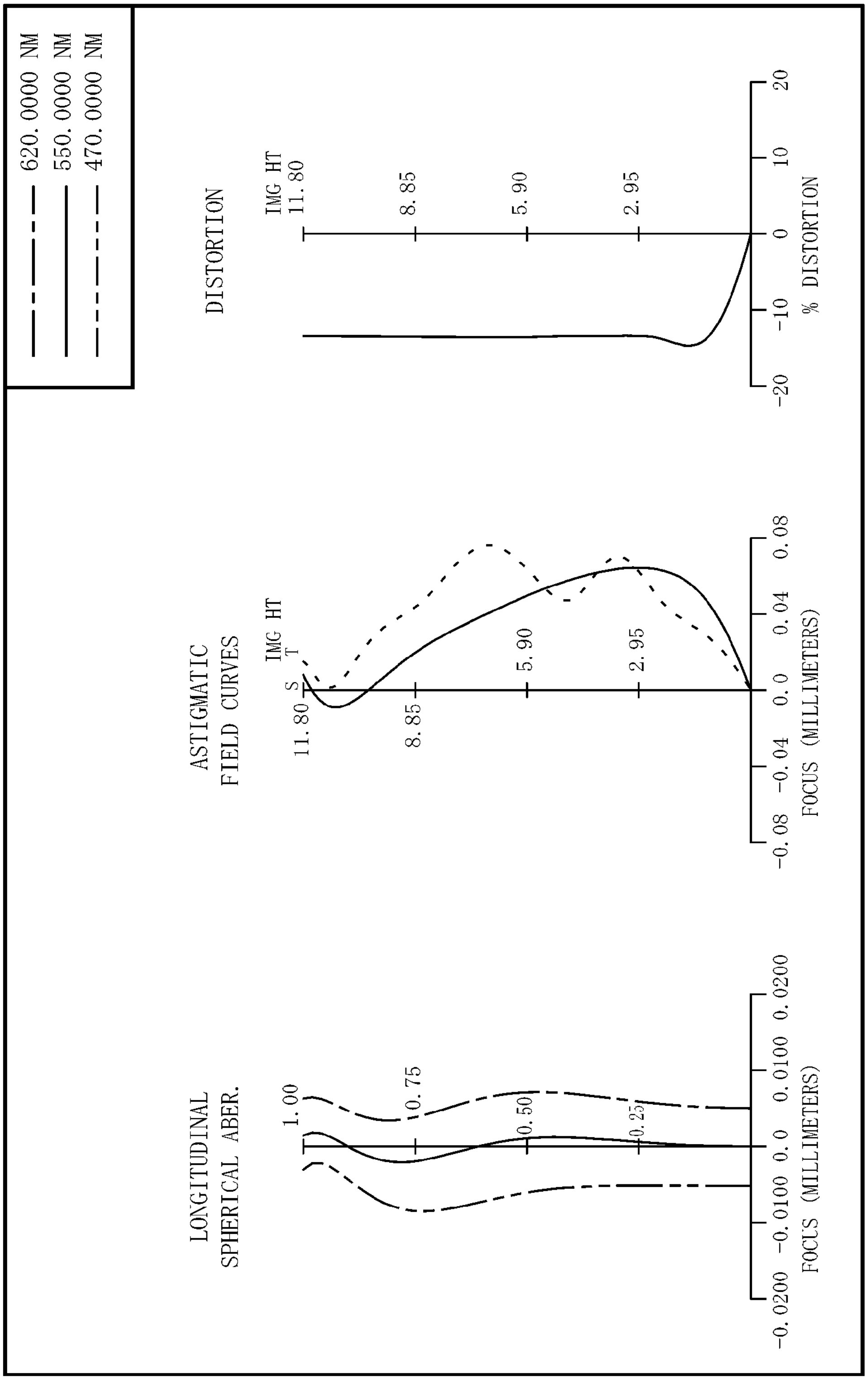
8.85
5.90
2.95
-20
-10
0
10
20
% DISTORTION

FIG. 10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8.85
5.90
2.95
S
T
-0.08 -0.04 0.0 0.04 0.08
FOCUS (MILLIMETERS)
DISTORTION
IMG HT
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5.90
2.95
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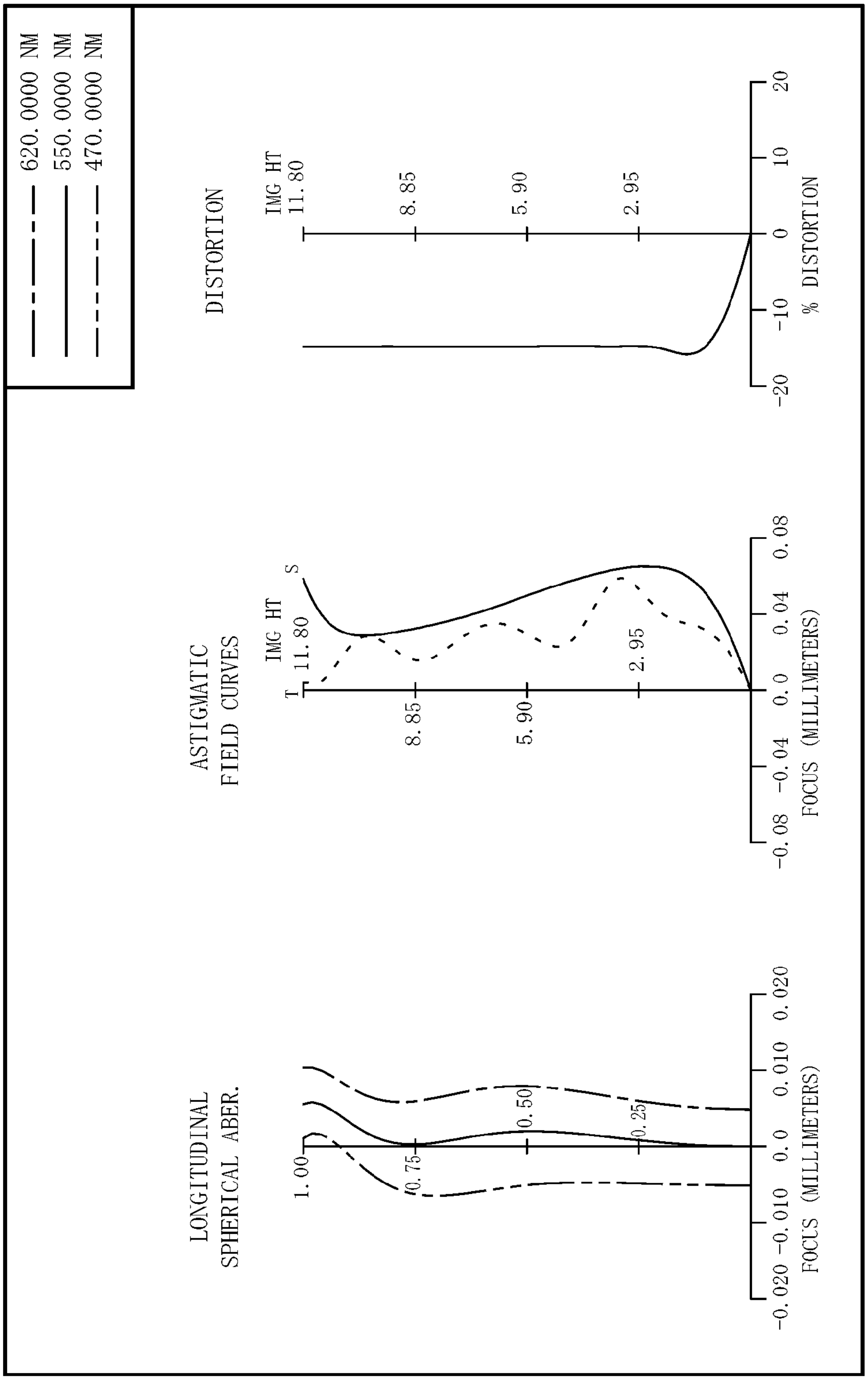
% DISTORTION

FIG. 11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ASTIGMATIC FIELD CURVES
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S
T
8.85
5.90
2.95
−0.08 −0.04 0.0 0.04 0.08
FOCUS (MILLIMETERS)
DISTORTION
IMG HT 11.80
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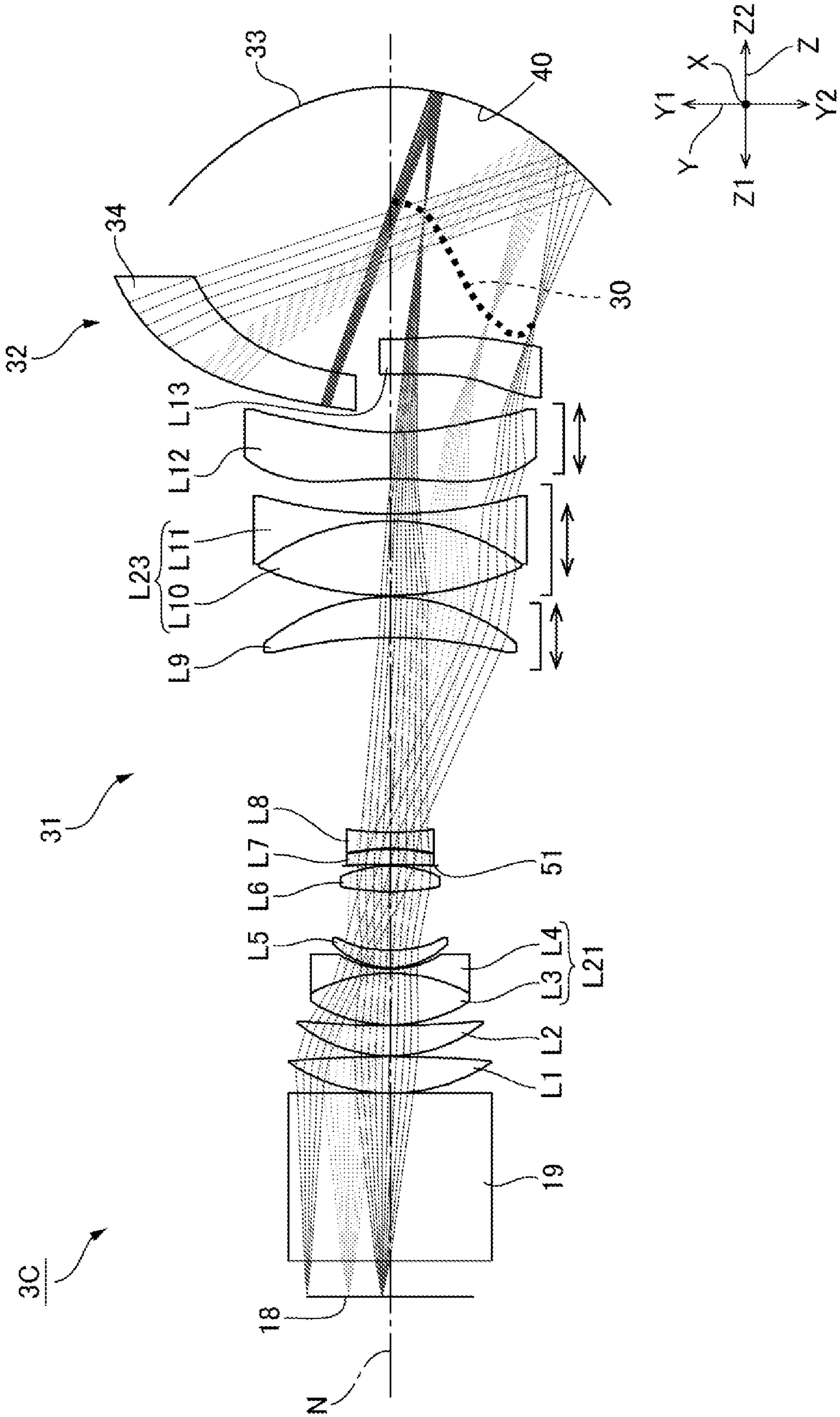
% DISTORTION

FIG. 12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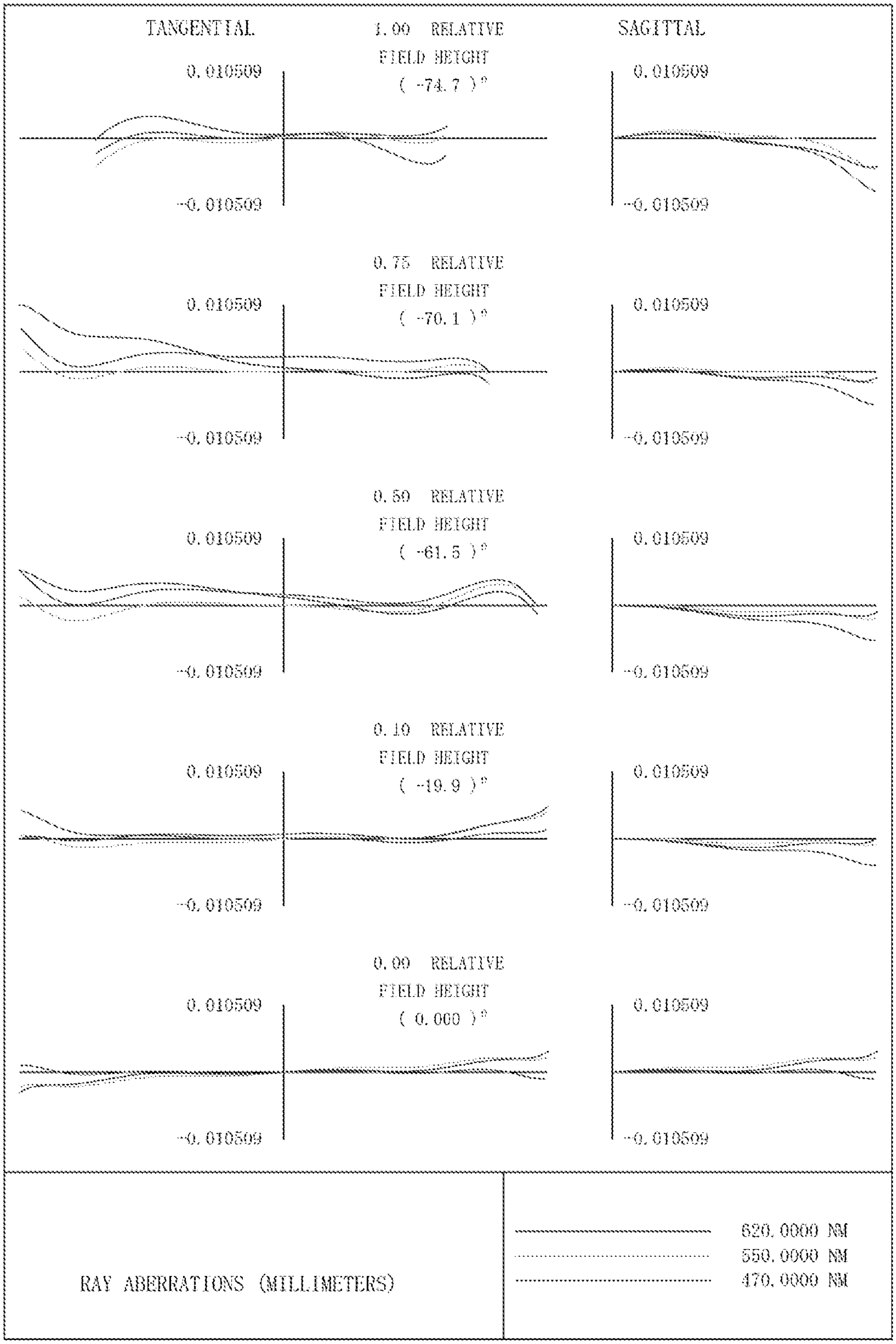
L21
51
19
18
N
X
Y
Y1
Y2
Z
Z1
Z2

FIG. 13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FIELD HEIGHT
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FIELD HEIGHT
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0.00 RELATIVE
FIELD HEIGHT
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RAY ABERRATIONS (MILLIMETERS)
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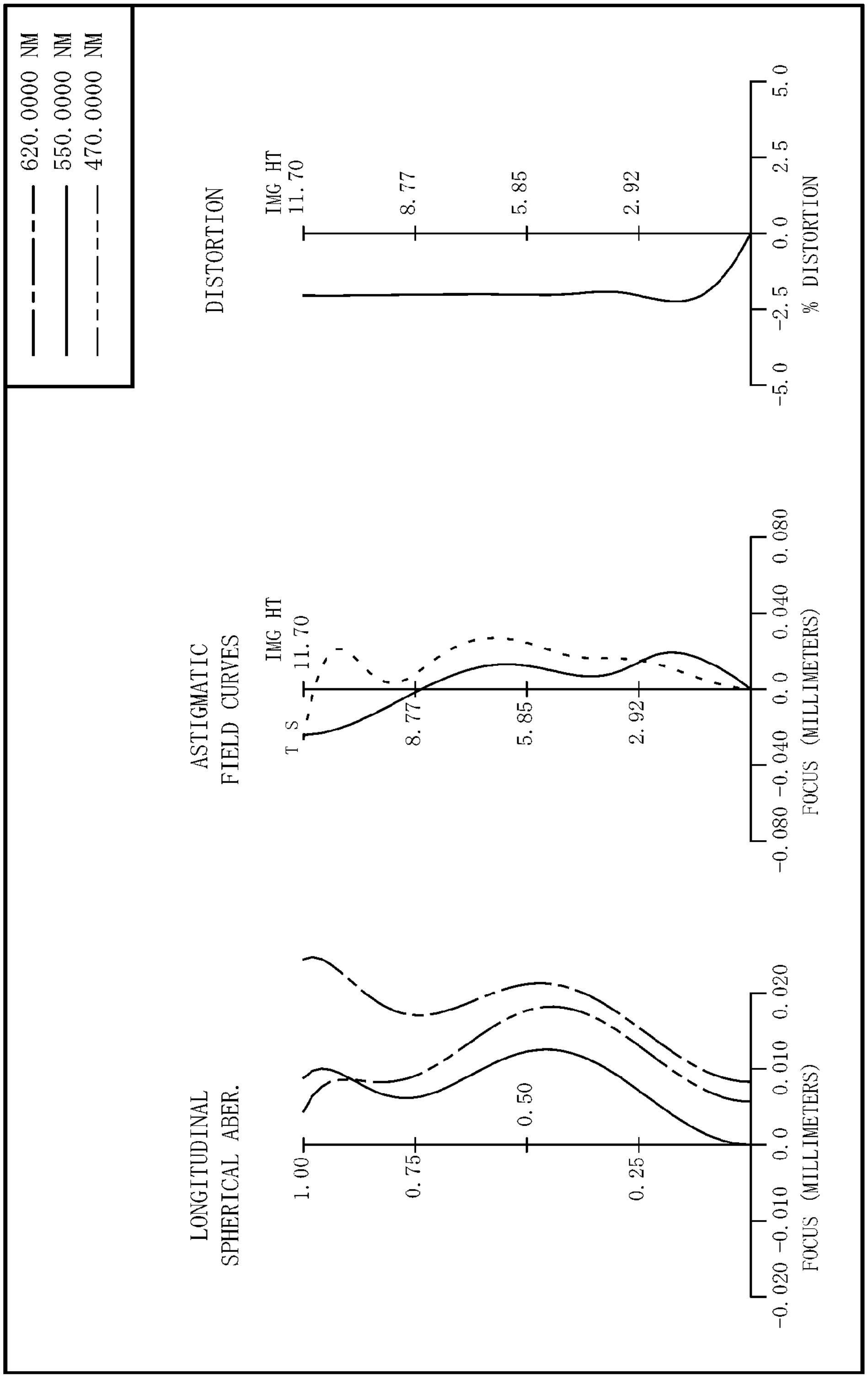
470.0000 NM

FIG. 14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FOCUS (MILLIMETERS)
ASTIGMATIC FIELD CURVES
IMG HT
11.70
T S
8.77
5.85
2.92
-0.080 -0.040 0.0 0.040 0.080
FOCUS (MILLIMETERS)
DISTORTION
IMG HT
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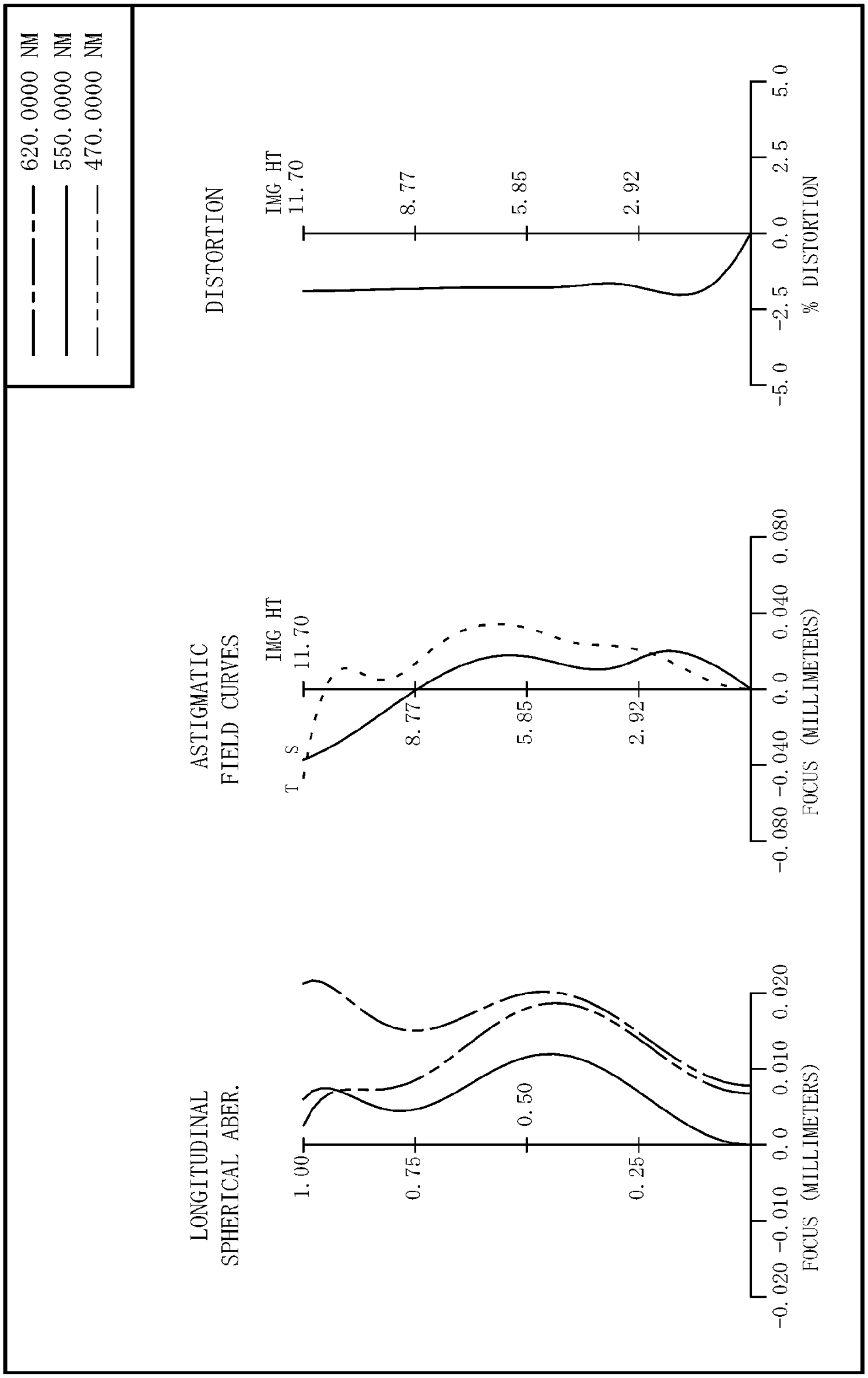
8.77
5.85
2.92
-5.0 -2.5 0.0 2.5 5.0
% DISTORTION

FIG. 15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SPHERICAL ABER.
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0.25
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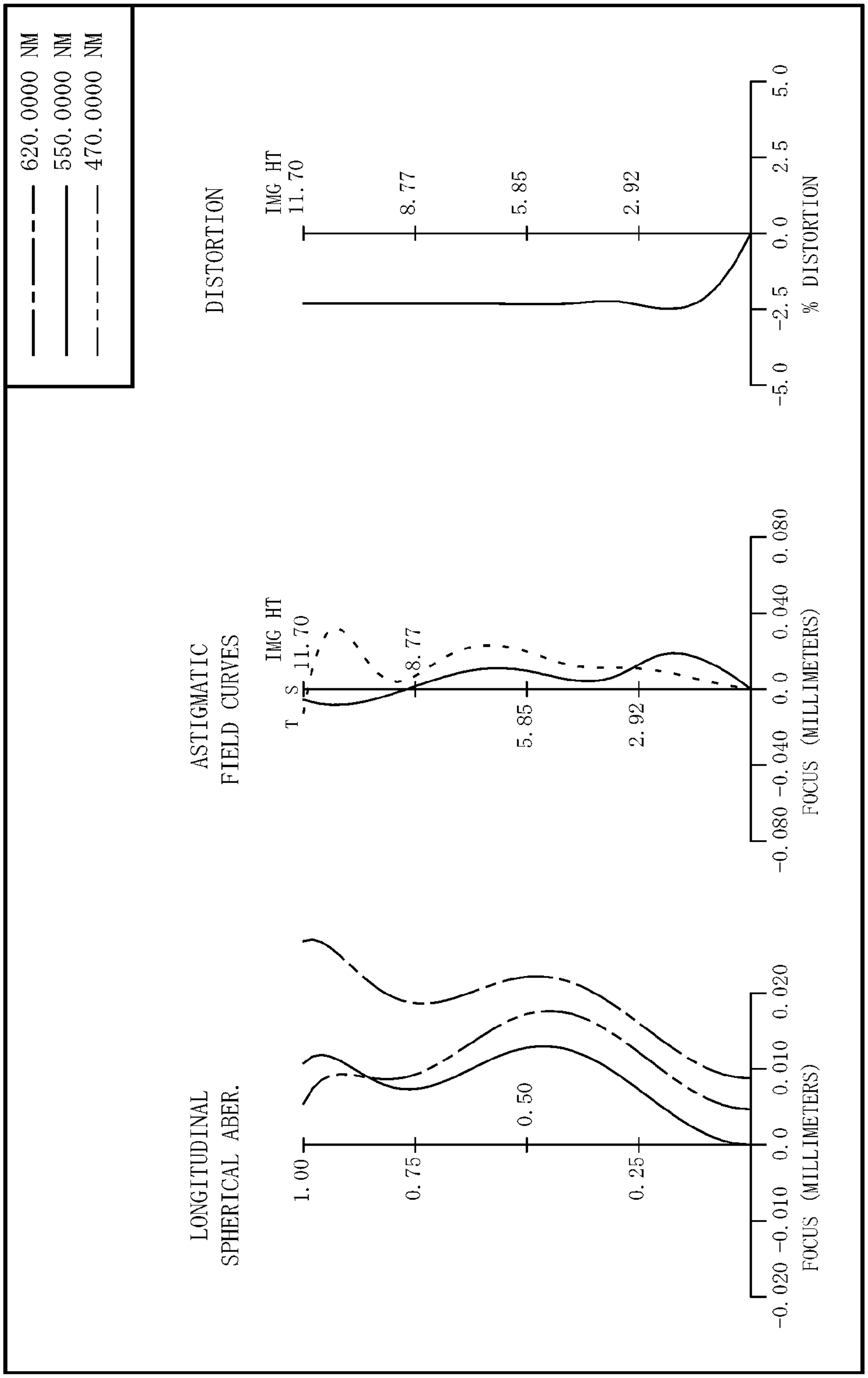
FOCUS (MILLIMETERS)
ASTIGMATIC
FIELD CURVES
IMG HT
11.70
T S
8.77
5.85
2.92
-0.080 -0.040 0.0 0.040 0.080
FOCUS (MILLIMETERS)
DISTORTION
IMG HT
11.70
8.77
5.85
2.92
-5.0 -2.5 0.0 2.5 5.0
% DISTORTION

FIG. 16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SPHERICAL ABER.
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FOCUS (MILLIMETERS)
ASTIGMATIC
FIELD CURVES
IMG HT
11.70
8.77
5.85
2.92
T
S
-0.080 -0.040 0.0 0.040 0.080
FOCUS (MILLIMETERS)
DISTORTION
IMG HT
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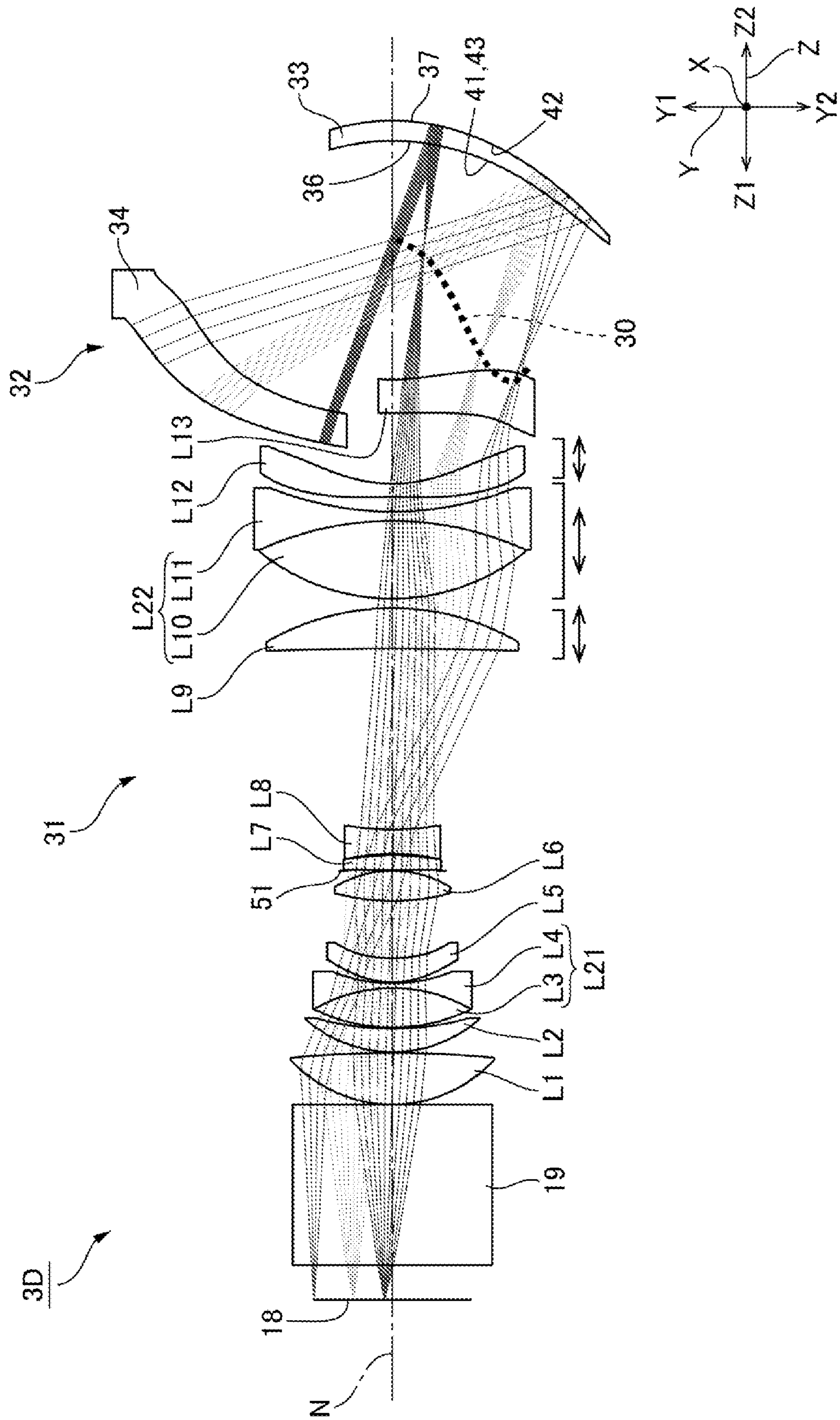
8.77
5.85
2.92
-5.0 -2.5 0.0 2.5 5.0
% DISTORTION

FIG. 17



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L1
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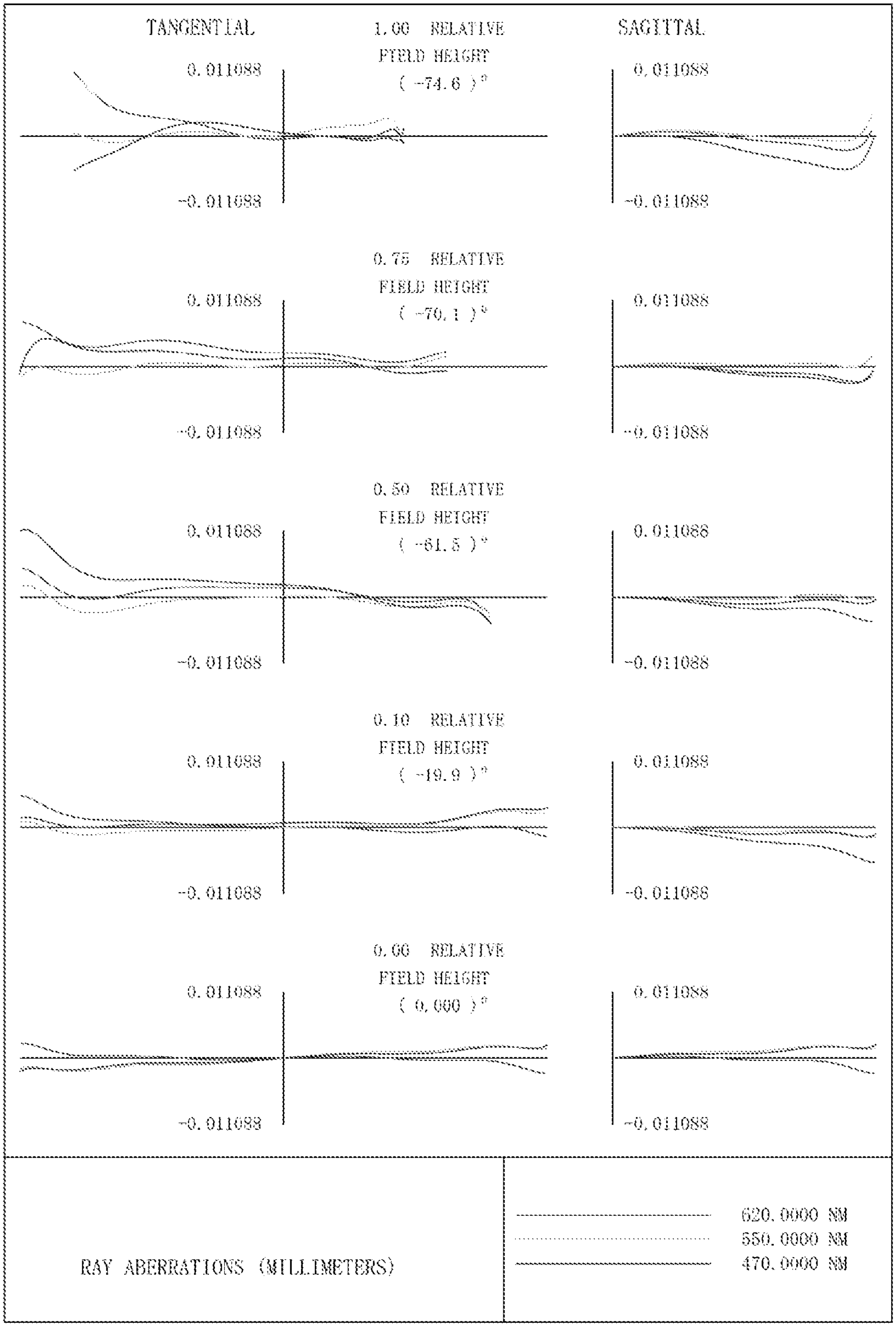
3D
18
N

FIG. 18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FIELD HEIGHT
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0.50 RELATIVE
FIELD HEIGHT
(-61.5)°
0.10 RELATIVE
FIELD HEIGHT
(-19.9)°
0.00 RELATIVE
FIELD HEIGHT
(0.000)°
RAY ABERRATIONS (MILLIMETERS)
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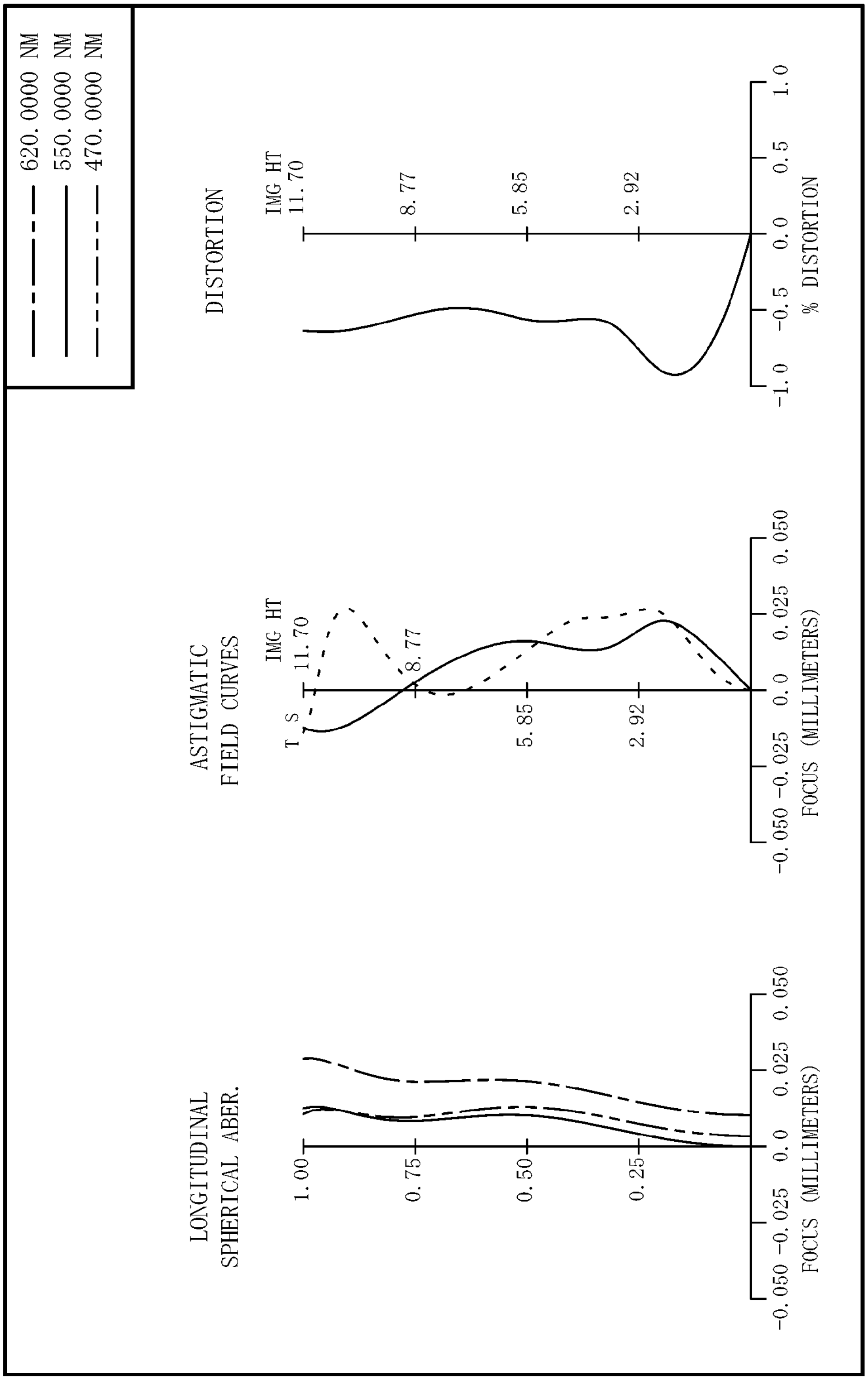
470.0000 NM

FIG. 19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550.0000 NM
470.0000 NM
LONGITUDINAL
SPHERICAL ABER.
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0.50
0.25
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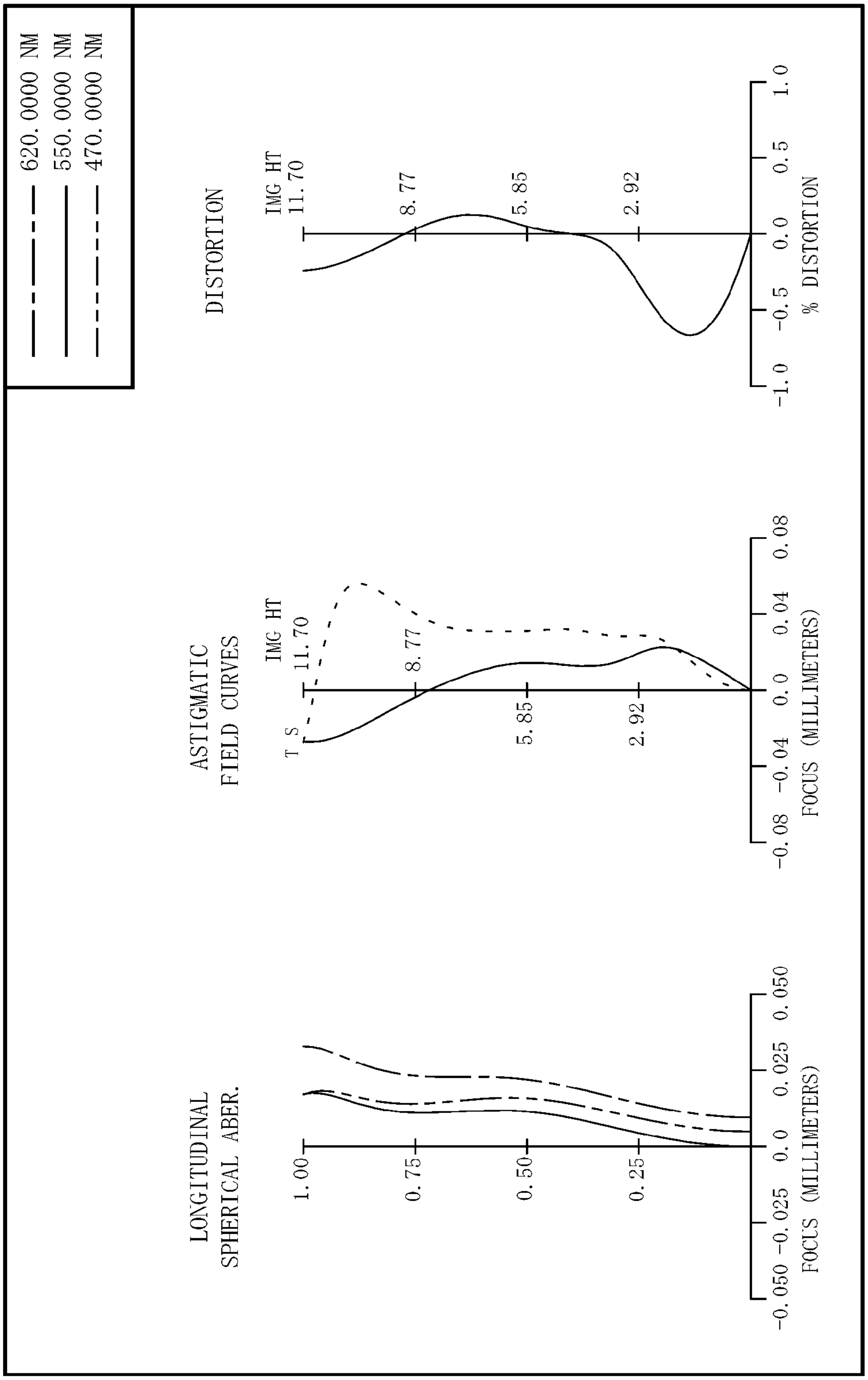
FOCUS (MILLIMETERS)
ASTIGMATIC
FIELD CURVES
IMG HT
11.70
8.77
5.85
2.92
T S
-0.050 -0.025 0.0 0.025 0.050
FOCUS (MILLIMETERS)
DISTORTION
IMG HT
11.70
8.77
5.85
2.92
-1.0 -0.5 0.0 0.5 1.0
% DISTORTION

FIG. 20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0.25
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FOCUS (MILLIMETERS)
ASTIGMATIC FIELD CURVES
IMG HT 11.70
8.77
5.85
2.92
T S
-0.08 -0.04 0.0 0.04 0.08
FOCUS (MILLIMETERS)
DISTORTION
IMG HT 11.70
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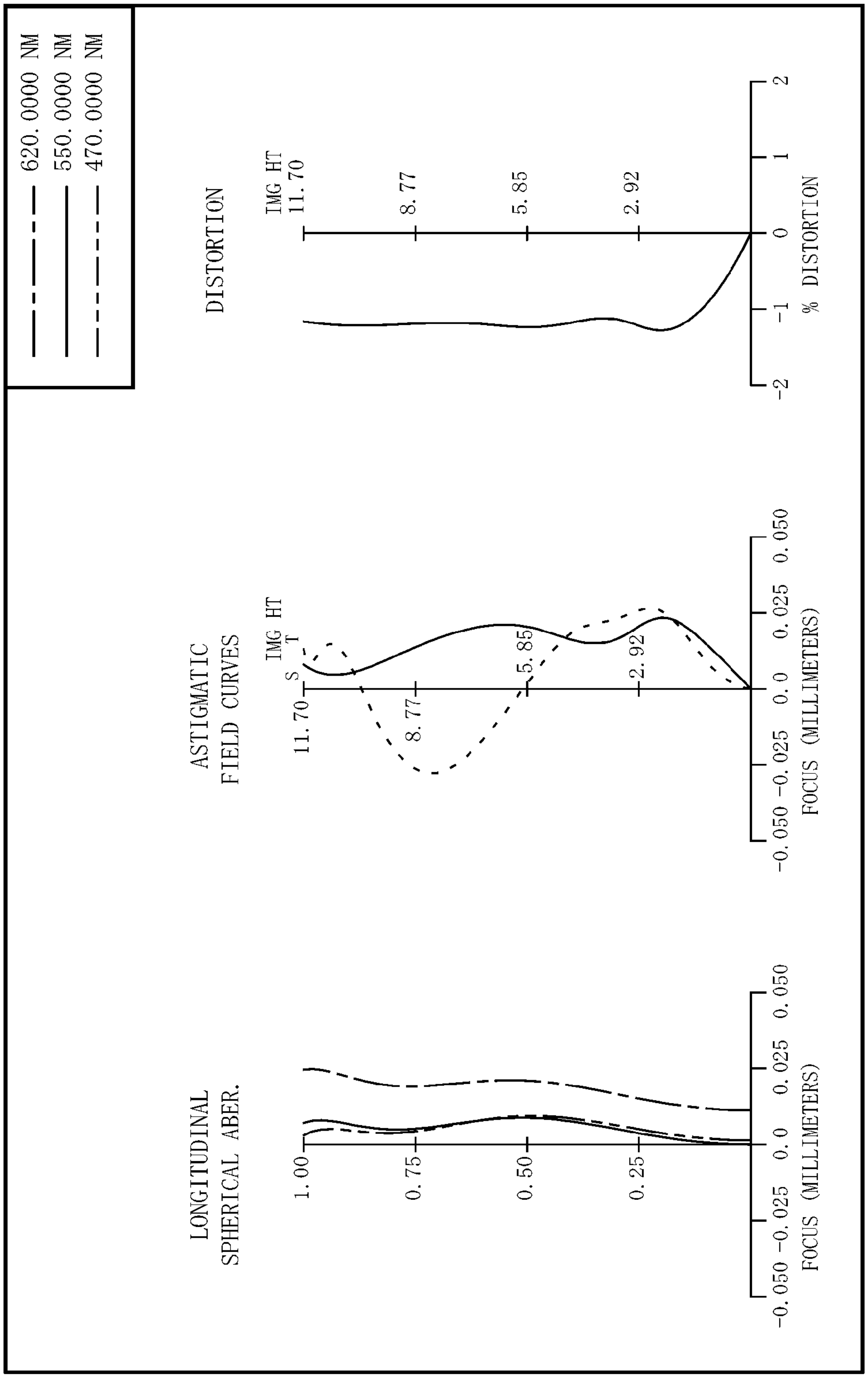
5.85
2.92
-1.0 -0.5 0.0 0.5 1.0
% DISTORTION

FIG. 21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SPHERICAL ABER.
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0.50
0.25
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FOCUS (MILLIMETERS)
ASTIGMATIC
FIELD CURVES
IMG HT
S
T
11.70
8.77
5.85
2.92
−0.050 −0.025 0.0 0.025 0.050
FOCUS (MILLIMETERS)
DISTORTION
IMG HT
11.70
8.77
5.85
2.92
−2 −1 0 1 2
% DISTORTION

PROJECTION SYSTEM AND PROJECTOR

The present application is based on, and claims priority from JP Application Serial Number 2022-006175, filed Jan. 19, 2022, the disclosure of which is hereby incorporated by reference herein in its entirety.

BACKGROUND

1. Technical Field

The present disclosure relates to a projection system and a projector.

2. Related Art

JP-A-2020-34690 describes a projector in which a projection system enlarges a projection image displayed at an image display device and projects the enlarged projection image onto a screen. The projection system includes a first refractive optical system, a reflective optical system, and a second refractive optical system sequentially arranged from the reduction side toward the enlargement side. The first refractive optical system includes a plurality of refractive lenses. The reflective optical system includes a concave mirror and reflects beams from the first refractive optical system toward the side facing the image display device in directions that intersect with the optical axis of the first refractive optical system. The second refractive optical system is formed of a single refractive lens. The refractive lens is an enlargement-side lens located at a position closest to the enlargement side in the projection system. Beams from the concave mirror enter the enlargement-side lens in directions that intersect with the optical axis of the enlargement-side lens.

Out of the examples of the projection system disclosed in JP-A-2020-34690, the projection system having the shortest projection distance has a projection distance of 257.6 mm. The enlargement-side lens of the thus configured projection system has an effective radius of 79.7 mm. The thus configured projection system further has a throw ratio of 0.154.

A projector including a projection system having a smaller throw ratio has a shorter projection distance over which the projector projects an enlarged image having a predetermined size. A projection system incorporated in a projector used indoors or at similar locations therefore needs to have a short focal length that provides a throw ratio smaller than or equal to 0.3.

A projection system having a shorter focal length tends to produce larger amounts of aberrations at the enlargement side. It is therefore necessary to increase the effective radius of the enlargement-side lens, through which the beams from the concave mirror obliquely pass, to allow the enlargement-side lens to correct the beams on an image height basis. When the size of the enlargement-side lens is increased to provide a sufficient effective radius, however, the amount of protrusion by which the enlargement-side lens protrudes radially from the first optical axis of the first refractive optical system increases, resulting in an increase in the diameter of the entire projection system. The size of the projector that incorporates the projection system is therefore not reduced.

SUMMARY

To solve the problem described above, a projection system according to an aspect of the present disclosure is a projection system for enlarging a projection image formed by an image formation device disposed in a reduction-side conjugate plane and projecting the enlarged image in an enlargement-side conjugate plane. The projection system includes a first optical system and a second optical system sequentially arranged from the reduction side toward the enlargement side. The first optical system includes a diaphragm. The second optical system includes an optical element having a concave reflection surface and a first lens having negative power, the optical element and the first lens sequentially arranged from the reduction side toward the enlargement side. An intermediate image conjugate with the reduction-side conjugate plane and the enlargement-side conjugate plane is formed between the first optical system and the second optical system. A portion at the reduction side of the first optical system forms a telecentric portion. The projection system satisfies all Conditional Expressions (1) and (2) below,

$$TR \leq 0.3 \tag{1}$$

$$35 \leq (OAL/imy) \times (LL/imy) \times TR \times (1/NA) \leq 60 \tag{2}$$

where OAL represents an axial inter-surface spacing from the image formation device to the reflection surface, imy represents a first distance from an optical axis to a largest image height at the image formation device, LL represents a largest radius of the first lens, TR represents a throw ratio that is a quotient of division of a projection distance by a second distance from the optical axis to a largest image height of the enlarged image, and NA represents a numerical aperture of the image formation device.

A projector according to another aspect of the present disclosure includes the projection system described above and the image formation device that forms a projection image in the reduction-side conjugate plane of the projection system.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. **1** shows a schematic configuration of a projector including a projection system according to an embodiment of the present disclosure.

FIG. **2** is a beam diagram showing beams passing through the projection system according to Example 1.

FIG. **3** shows lateral aberrations produced by the projection system according to Example 1 set at a standard distance.

FIG. **4** shows spherical aberration, astigmatism, and distortion produced by the projection system according to Example 1 set at the standard distance.

FIG. **5** shows the spherical aberration, astigmatism, and distortion produced by the projection system according to Example 1 set at a short distance.

FIG. **6** shows the spherical aberration, astigmatism, and distortion produced by the projection system according to Example 1 set at a long distance.

FIG. **7** is a beam diagram showing beams passing through the projection system according to Example 2.

FIG. **8** shows the lateral aberrations produced by the projection system according to Example 2 set at the standard distance.

FIG. **9** shows the spherical aberration, astigmatism, and distortion produced by the projection system according to Example 2 set at the standard distance.

FIG. **10** shows the spherical aberration, astigmatism, and distortion produced by the projection system according to Example 2 set at the short distance.

FIG. **11** shows the spherical aberration, astigmatism, and distortion produced by the projection system according to Example 2 set at the long distance.

FIG. **12** is a beam diagram showing beams passing through the projection system according to Example 3.

FIG. **13** shows the lateral aberrations produced by the projection system according to Example 3 set at the standard distance.

FIG. **14** shows the spherical aberration, astigmatism, and distortion produced by the projection system according to Example 3 set at the standard distance.

FIG. **15** shows the spherical aberration, astigmatism, and distortion produced by the projection system according to Example 3 set at the short distance.

FIG. **16** shows the spherical aberration, astigmatism, and distortion produced by the projection system according to Example 3 set at the long distance.

FIG. **17** is a beam diagram showing beams passing through the projection system according to Example 4.

FIG. **18** shows the lateral aberrations produced by the projection system according to Example 4 set at the standard distance.

FIG. **19** shows the spherical aberration, astigmatism, and distortion produced by the projection system according to Example 4 set at the standard distance.

FIG. **20** shows the spherical aberration, astigmatism, and distortion produced by the projection system according to Example 4 set at the short distance.

FIG. **21** shows the spherical aberration, astigmatism, and distortion produced by the projection system according to Example 4 set at the long distance.

DESCRIPTION OF EXEMPLARY EMBODIMENTS

An optical system and a projector according to an embodiment of the present disclosure will be described below with reference to the drawings.

Projector

FIG. **1** shows a schematic configuration of a projector including a projection system **3** according to the embodiment of the present disclosure. A projector **1** includes an image formation unit **2**, which generates a projection image to be projected onto a screen S, the projection system **3**, which enlarges the projection image and projects the enlarged image onto the screen S, and a controller **4**, which controls the operation of the image formation unit **2**, as shown in FIG. **1**.

Image Formation Unit and Controller

The image formation unit **2** includes a light source **10**, a first optical integration lens **11**, a second optical integration lens **12**, a polarization converter **13**, and a superimposing lens **14**. The light source **10** is formed, for example, of an ultrahigh-pressure mercury lamp or a solid-state light source. The first optical integration lens **11** and the second optical integration lens **12** each include a plurality of lens elements arranged in an array. The first optical integration lens **11** divides a luminous flux from the light source **10** into a plurality of luminous fluxes. The lens elements of the first optical integration lens **11** focus the luminous flux from the light source **10** in the vicinity of the lens elements of the second optical integration lens **12**.

The polarization converter **13** converts the light from the second optical integration lens **12** into predetermined linearly polarized light. The superimposing lens **14** superimposes images of the lens elements of the first optical integration lens **11** on one another in a display area of each of liquid crystal panels **18**R, **18**G, and **18**B, which will be described later, via the second optical integration lens **12**.

The image formation unit **2** further includes a first dichroic mirror **15**, a reflection mirror **16**, a field lens **17**R, and the liquid crystal panel **18**R. The first dichroic mirror **15** reflects R light, which is part of the beams incident via the superimposing lens **14**, and transmits G light and B light, which are part of the beams incident via the superimposing lens **14**. The R light reflected off the first dichroic mirror **15** travels via the reflection mirror **16** and the field lens **17**R and is incident on the liquid crystal panel **18**R. The liquid crystal panel **18**R is an image formation element. The liquid crystal panel **18**R modulates the R light in accordance with an image signal to form a red projection image.

The image formation unit **2** further includes a second dichroic mirror **21**, a field lens **17**G, and the liquid crystal panel **18**G. The second dichroic mirror **21** reflects the G light, which is part of the beams via the first dichroic mirror **15**, and transmits the B light, which is part of the beams via the first dichroic mirror **15**. The G light reflected off the second dichroic mirror **21** passes through the field lens **17**G and is incident on the liquid crystal panel **18**G. The liquid crystal panel **18**G is an image formation element. The liquid crystal panel **18**G modulates the G light in accordance with an image signal to form a green projection image.

The image formation unit **2** further includes a relay lens **22**, a reflection mirror **23**, a relay lens **24**, a reflection mirror **25**, a field lens **17**B, the liquid crystal panel **18**B, and a cross dichroic prism **19**. The B light having passed through the second dichroic mirror **21** travels via the relay lens **22**, the reflection mirror **23**, the relay lens **24**, the reflection mirror **25**, and the field lens **17**B and is incident on the liquid crystal panel **18**B. The liquid crystal panel **18**B is an image formation element. The liquid crystal panel **18**B modulates the B light in accordance with an image signal to form a blue projection image.

The liquid crystal panels **18**R, **18**G, and **18**B surround the cross dichroic prism **19** so as to face three sides of the cross dichroic prism **19**. The cross dichroic prism **19** is a prism for light combination and produces a projection image that is the combination of the light modulated by the liquid crystal panel **18**R, the light modulated by the liquid crystal panel **18**G, and the light modulated by the liquid crystal panel **18**B.

The projection system **3** enlarges the combined projection image from the cross dichroic prism **19** and projects the enlarged projection image onto the screen S.

The control unit **4** includes an image processor **6**, to which an external image signal, such as a video signal, is inputted, and a display driver **7**, which drives the liquid crystal panels **18**R, **18**G, and **18**B based on image signals outputted from the image processor **6**.

The image processor **6** converts the image signal inputted from an external apparatus into image signals each containing grayscales and other factors of a color corresponding to the image signal. The display driver **7** operates the liquid crystal panels **18**R, **18**G, and **18**B based on the color projection image signals outputted from the image processor **6**. The image processor **6** thus causes the liquid crystal panels **18**R, **18**G, and **18**B to display projection images corresponding to the image signals.

Projection System

The projection system **3** will next be described. The screen S is disposed in the enlargement-side conjugate plane of the projection system **3**, as shown in FIG. **1**. The liquid crystal panels **18**R, **18**G, and **18**B are disposed in the reduction-side conjugate plane of the projection system **3**.

Examples 1 to 4 will be described below as examples of the configuration of the projection system **3** incorporated in the projector **1**.

Example 1

FIG. **2** is a beam diagram showing beams passing through a projection system **3**A according to Example 1. In the beam diagrams for the projection systems **3** according to Examples 1 to 4, the liquid crystal panels **18**R, **18**G, and **18**B are referred to as a liquid crystal panel **18**. The projection system **3**A according to the present example is formed of a first optical system **31** and a second optical system **32** sequentially arranged from the reduction side toward the enlargement side, as shown in FIG. **2**. The second optical system **32** is disposed on an optical axis N of the first optical system **31**.

In the following description, three axes perpendicular to one another are called axes X, Y, and Z for convenience. The axis Z coincides with the optical axis N of the first optical system **31**. The direction along the optical axis N is an axis-Z direction. The axis-Z direction toward the side where the first optical system **31** is located is called a first direction Z**1**, and the axis-Z direction toward the side where the second optical system **32** is located is called a second direction Z**2**. The axis Y extends along the screen S. The upward-downward direction is an axis-Y direction, with one side of the axis-Y direction called an upper side Y**1** and the other side of the axis-Y direction called a lower side Y**2**. The axis X extends in the width direction of the screen.

The first optical system **31** is a refractive optical system. The first optical system **31** is formed of 17 lenses L**1** to L**17**. The lenses L**1** to L**17** are arranged in the presented order from the reduction side toward the enlargement side. A diaphragm **51** is disposed between the lens L**7** and the lens L**8**.

The lens L**6** has aspherical shapes at opposite sides. The lens L**9** has aspherical shapes at opposite sides. The lens L**16** (third lens) has aspherical shapes at opposite sides. The lens L**17** (second lens) has aspherical shapes at opposite sides. The lens L**2** and the lens L**3** are bonded to each other into a cemented doublet L**21**. The lens L**4** and the lens L**5** are bonded to each other into a cemented doublet L**22**. The lens L**11** and the lens L**12** are bonded to each other into a cemented doublet L**23**. The lens L**14** and the lens L**15** are bonded to each other into a cemented doublet L**24**.

The second optical system **32** includes an optical element **33** and a first lens **34**. The optical element **33** and the first lens **34** are arranged in the presented order from the reduction side toward the enlargement side. The optical element **33** has a reflection surface **40**, which faces the reduction side. The reflection surface **40** has a concave shape recessed in the second direction Z**2**. The reflection surface **40** has an aspherical shape. The reflection surface **40** is located at the lower side Y**2** of the optical axis N, as shown in FIG. **2**. The reflection surface **40** is formed by providing the outer surface, in the first direction Z**1**, of the optical element **33** with a reflection coating layer (reflection layer). The reflection surface **40** reflects light at the surface, facing in the direction Z**1**, of the optical element **33**.

The first lens **34** is located at a position shifted from the optical element **33** in the first direction Z**1** and disposed at the upper side Y**1** of the optical axis N. The first lens **34** has negative power. The first lens **34** has a convex enlargement-side surface and a concave reduction-side surface. The first lens **34** has aspherical shapes at opposite sides.

The liquid crystal panel **18** of the image formation unit **2** is disposed in the reduction-side conjugate plane of the projection system **3**A. The screen S is disposed in the enlargement-side conjugate plane of the projection system **3**A.

The liquid crystal panel **18** forms a projection image in an image formation plane perpendicular to the optical axis N of the first optical system **31**. The liquid crystal panel **18** is disposed in a position offset from the optical axis N of the first optical system **31** toward the upper side Y**1**. The projection image is therefore formed in a position offset from the optical axis N toward the upper side Y**1**.

The beams from the liquid crystal panel **18** pass through the first optical system **31** and the second optical system **32** in the presented order. Between the first optical system **31** and the second optical system **32**, the beams pass through the lower side Y**2** of the optical axis N. The beams are therefore directed through the second optical system **32** toward the reflection surface **40**. The beams having reached the reflection surface **40** are deflected back in the first direction Z**1** towards the upper side Y**1**. The beams deflected back by the reflection surface **40** cross the optical axis N toward the upper side Y**1** and travels toward the first lens **34**. The beams passing through the first lens **34** are widened by the first lens **34** and reach the screen S.

The lens L**17** of the first optical system **31** is disposed between the reflection surface **40** and the first lens **34** in the direction of the optical axis N. An intermediate image **30** is formed between the lens L**17** and the reflection surface **40**.

In the projection system **3**A, the portion at the reduction side of the first optical system **31** is a telecentric portion. The term "telecentric" means that the central beam of each luminous flux traveling between the first optical system **31** and the liquid crystal panel **18** disposed in the reduction-side conjugate plane is parallel or substantially parallel to the optical axis of the projection system.

The projection system **3**A has a changeable projection distance. When the projection distance is changed, eight lenses of the first optical system **31**, the lenses L**10** to L**17**, are moved along the optical axis N for focusing. In the focusing, the lenses L**10**, L**11**, and L**12** are moved as a unit. In the focusing, the lenses L**13**, L**14**, and L**15** are moved also as a unit.

Data on the projection system **3**A are listed below,

NA	0.3125
imy	11.7 mm
scy	1463 mm
PD	283.1 mm
M	125
TR	0.194
OAL	203 mm
LL	47.8 mm

where NA represents the numerical aperture of the liquid crystal panel **18**, imy represents a first distance from the optical axis N to the largest image height at the liquid crystal panel **18**, scy represents a second distance from the optical axis N to the largest image height of the enlarged image projected on the screen S, PD represents a projection distance that is the distance from the first lens **34** to the screen S, M represents a projection magnification that is the quotient of division of the second distance by the first distance, TR represents a throw ratio that is the quotient of division of the projection distance by the second distance, OAL represents the axial inter-surface spacing from the liquid crystal panel **18** to the reflection surface **40**, and LL represents the largest radius of the first lens **34**.

Data on the lenses of the projection system **3**A are listed below. The surfaces of the lenses are numbered sequentially from the reduction side toward the enlargement side. Reference characters are given to the liquid crystal panel, the dichroic prism, the lenses, the optical element, the first lens, and the screen. Data labeled with a surface number that does not correspond to any of the liquid crystal panel, the dichroic prism, the lenses, the optical element, the first lens, and the screen is dummy data. Reference character R represents the radius of curvature. Reference character D represents the axial inter-surface spacing. Reference character C represents the aperture radius, and twice the aperture radius is the diameter of the lens surface. Reference characters R, D, and C are each expressed in millimeters.

Reference character	Surface number	Shape	R	D	Glass material	Refraction/ Reflection	C
18	0	Spherical	Infinity	9.5000		Refraction	0.0000
19	1	Spherical	Infinity	25.9100	SBSL7_OHARA	Refraction	13.0371
	2	Spherical	Infinity	0.0000		Refraction	15.4259
L1	3	Spherical	23.6460	8.4106	SFPL51_OHARA	Refraction	16.3497
	4	Spherical	−163.6273	0.1000		Refraction	16.1238
L2	5	Spherical	24.1479	5.7206	SFSL5_OHARA	Refraction	14.0000
L3	6	Spherical	113.0998	1.0000	STIH6_OHARA	Refraction	13.1404
	7	Spherical	34.7820	0.1000		Refraction	12.2395
L4	8	Spherical	19.4148	7.7264	SBSL7_OHARA	Refraction	11.6173
L5	9	Spherical	−25.1932	0.9500	TAFD25_HOYA	Refraction	10.9949
	10	Spherical	25.2418	0.2000		Refraction	10.0083
L6	11	Aspherical	16.8372	4.4033	LBAL35_OHARA	Refraction	10.1020
	12	Aspherical	40.5023	1.0000		Refraction	9.1836
L7	13	Spherical	19.4289	2.3917	SFSL5_OHARA	Refraction	9.1301
	14	Spherical	32.7745	4.0203		Refraction	8.8594
51	15	Spherical	Infinity	0.1043		Refraction	8.3951
L8	16	Spherical	101.2882	3.7504	STIH53_OHARA	Refraction	8.5137
	17	Spherical	−22.7544	0.1000		Refraction	8.6718
L9	18	Aspherical	−21.2771	6.0000	LLAM60_OHARA	Refraction	8.6423
	19	Aspherical	103.2802	Variable spacing 1		Refraction	9.5014
L10	20	Spherical	22.9132	2.0346	STIM22_OHARA	Refraction	10.9384
	21	Spherical	26.8895	14.2089		Refraction	10.9386
L11	22	Spherical	57.1566	5.3002	STIM2_OHARA	Refraction	16.5000
L12	23	Spherical	−129.3659	1.0000	STIH6_OHARA	Refraction	16.7439
	24	Spherical	677.5558	Variable spacing 2		Refraction	17.0582
L13	25	Spherical	47.4289	9.6511	STIM22_OHARA	Refraction	18.3999
	26	Spherical	−46.5818	0.1000		Refraction	18.3651
L14	27	Spherical	−67.5767	8.1053	STIL25_OHARA	Refraction	17.8500
L15	28	Spherical	−21.6948	1.0000	STIH6_OHARA	Refraction	17.7177
	29	Spherical	170.7327	Variable spacing 3		Refraction	18.7445
L16	30	Aspherical	−24.7550	3.0000	'Z-E48R'	Refraction	18.8159
	31	Aspherical	101.4895	Variable spacing 4		Refraction	21.3950
L17	32	Aspherical	257.2804	8.0000	'Z-E48R'	Refraction	25.6568
	33	Aspherical	63.8637	Variable spacing 5		Refraction	28.1976
40	34	Aspherical	−28.5800	−50.8918		Reflection	
34	35	Aspherical	59.7480	−7.0000	'Z-E48R'	Refraction	37.2465
	36	Aspherical	67.7207	Variable spacing 6		Refraction	47.7801
S	37	Spherical	Infinity	0.0000		Refraction	1985.1150

The projection system **3**A according to the present example has a changeable projection distance selected from a standard distance, a short distance shorter than the standard distance, and a long distance longer than the standard distance. When the projection distance is changed, eight lenses of the first optical system **31**, the lenses L**10** to L**17**, are moved along the optical axis N for focusing. When the focusing is performed so as to change the projection distance from the short distance to the long distance, the lenses L**10**, L**11**, and L**12** move along the optical axis N toward the enlargement side. In the same focusing operation, the lenses L**13**, L**14**, and L**15** move along the optical axis N toward the enlargement side. In the same focusing operation, the lens L**16** moves along the optical axis N toward the enlargement side. In the same focusing operation, the lens L**17** moves along the optical axis N toward the reduction side.

The table below shows the variable spacings 1, 2, 3, 4, 5, and 6 at the projection distances where the focusing is performed. The variable spacing 1 is the axial inter-surface spacing between the lens L**9** and the lens L**10**. The variable spacing 2 is the axial inter-surface spacing between the lens L**12** and the lens L**13**. The variable spacing 3 is the axial inter-surface spacing between the lens L**15** and the lens L**16**. The variable spacing 4 is the axial inter-surface spacing between the lens L**16** and the lens L**17**. The variable spacing 5 is the axial inter-surface spacing between the lens L**17** and the reflection surface **40**. The variable spacing 6 is the projection distance.

	Standard distance	Short distance	Long distance
Variable spacing 1	1.1069	0.8768	1.2979
Variable spacing 2	0.8042	0.1000	1.4634
Variable spacing 3	6.2530	6.4325	6.0906
Variable spacing 4	20.9176	21.6588	20.1541
Variable spacing 5	39.8875	39.8713	39.9334
Variable spacing 6	−283.0000	−217.0000	−401.0000

The aspherical coefficients are listed below.

Surface number	S11	S12	S18	S19
Radius of curvature (R)	16.8372	40.5023	−21.2771	103.2802
Conic constant (K)	8.76367E−01	1.40552E+01	−1	−90
Fourth-order	−2.95336E−06	8.91728E−05	4.09548E−05	7.14087E−05
Sixth-order	5.97014E−08	2.72315E−07	−4.12331E−07	−3.41490E−07
Eighth-order	7.38171E−11	−7.34670E−10	1.32697E−09	1.10129E−09
Tenth-order		1.42398E−11		

Surface number	S30	S31	S32	S33
Radius of curvature (R)	−24.7550	101.4895	257.2804	63.8637
Conic constant (K)	0	0	90	0.00000E+00
Fourth-order	9.42799E−05	1.68772E−05	−4.34174E−05	−6.79148E−05
Sixth-order	−3.84902E−07	−1.78130E−07	1.23752E−07	1.63921E−07
Eighth-order	8.80106E−10	3.67946E−10	−2.74764E−10	−2.83439E−10
Tenth-order	−8.42504E−13	−3.17825E−13	3.77647E−13	2.91903E−13
Twelfth-order			−2.03100E−16	−1.17046E−16
Fourteenth-order				6.56482E−22

Surface number	S34	S35	S36
Radius of curvature (R)	−28.5800	59.7480	67.72067982
Conic constant (K)	−1.00000E+00	2.18390E−01	−0.004229729
Fourth-order	3.62586E−06	2.99552E−05	1.47026E−05
Sixth-order	−6.05961E−09	−9.57606E−08	−3.80264E−08
Eighth-order	7.45065E−12	1.64656E−10	4.62847E−11
Tenth-order	−5.45419E−15	−1.61839E−13	−3.18524E−14
Twelfth-order	2.08318E−18	9.28635E−17	1.29283E−17
Fourteenth-order	−3.40484E−22	−2.85578E−20	−2.89715E−21
Sixteenth-order		3.61610E−24	2.80551E−25

The projection system **3**A according to the present example satisfies all Conditional Expressions (1) and (2) below,

$$TR \leq 0.3 \quad (1)$$

$$35 \leq (OAL/imy) \times (LL/imy) \times TR \times (1/NA) \leq 60 \quad (2)$$

where OAL represents the axial inter-surface spacing from the liquid crystal panel **18** to the reflection surface **40**, imy represents the first distance from the optical axis N to the largest image height at the liquid crystal panel **18**, LL represents the largest radius of the first lens **34**, TR represents the throw ratio, which is the quotient of division of the projection distance by the second distance from the optical axis N to the largest image height of the enlarged image projected on the screen S, and NA represents the numerical aperture of the liquid crystal panel **18**.

Furthermore, it is more preferable that the projection system **3**A satisfies all Conditional Expressions (1) and (2') below.

$$TR \leq 0.3 \quad (1)$$

$$35 \leq (OAL/imy) \times (LL/imy) \times TR \times (1/NA) \leq 53 \quad (2')$$

In the present example, the values described above are listed below.

OAL	203 mm
imy	11.7 mm
LL	47.8 mm
TR	0.194
NA	0.3125

TR=0.194 is provided from the table shown above, so that Conditional Expression (1) is satisfied. (OAL/imy)×(LL/imy)×TR×(1/NA)=44 is satisfied, so that Conditional Expression (2) is satisfied.

Effects and Advantages

The projection system **3**A according to the present example enlarges a projection image formed by the liquid crystal panel **18** disposed in the reduction-side conjugate plane and projects the enlarged projection image in the enlargement-side conjugate plane. The projection system **3**A according to the present example includes the first optical system **31** and the second optical system **32** sequentially arranged from the reduction side toward the enlargement side. The first optical system **31** includes the diaphragm **51**. The second optical system **32** includes the optical element **33**, which has the concave reflection surface **40**, and the first lens **34**, which has negative power, sequentially arranged from the reduction side toward the enlargement side. The intermediate image **30** conjugate with the reduction-side conjugate plane and the enlargement-side conjugate plane is formed between the first optical system **31** and the second optical system **32**. The portion at the reduction side of the first optical system **31** form a telecentric portion.

The projection system **3**A according to the present example satisfies all Conditional Expressions (1) and (2) below,

$$TR \leq 0.3 \tag{1}$$

$$35 \leq (OAL/imy) \times (LL/imy) \times TR \times (1/NA) \leq 60 \tag{2}$$

where OAL represents the axial inter-surface spacing from the liquid crystal panel **18** to the reflection surface **40**, imy represents the first distance from the optical axis N to the largest image height at the liquid crystal panel **18**, LL represents the largest radius of the first lens **34**, TR represents the throw ratio, which is the quotient of division of the projection distance by the second distance from the optical axis N to the largest image height of the enlarged image projected on the screen S, and NA represents the numerical aperture of the liquid crystal panel **18**.

The projection system **3**A according to the present example satisfies Conditional Expression (1). The projection system **3** therefore has a short focal length. A projection system having a shorter focal length tends to produce larger amounts of aberrations at the enlargement side. It is therefore necessary to increase the effective radius of the enlargement-side lens, through which the beams from the concave mirror obliquely pass, to allow the enlargement-side lens to correct the beams on an image height basis. When the size of the enlargement-side lens is increased to provide a sufficient effective radius, however, the amount of protrusion by which the enlargement-side lens protrudes radially from the first optical axis of the first refractive optical system increase, resulting in an increase in the diameter of the entire projection system.

To solve the problem described above, the projection system **3**A according to the present example satisfies Conditional Expression (2). Suppression of the amount of protrusion by which the first lens **34** protrudes radially from the optical axis N can therefore suppress an increase in the diameter of the entire projection system, whereby the size of the projector that incorporates the projection system **3**A can be reduced. Furthermore, the effective diameter of the first lens **34** within which the beams can be corrected on an image height basis can be ensured, while the amount of protrusion by which the first lens **34** protrudes radially from the optical axis N is suppressed. That is, when (OAL/imy)×(LL/imy)×TR×(1/NA) in Conditional Expression (2) is smaller than the lower limit, the axial inter-surface spacing from the liquid crystal panel **18** to the reflection surface **40** and the lens diameter of the first lens **34** become too small relative to TR and 1/NA, so that it is difficult to correct the beams on an image height basis, and sufficient resolution of the projection system **3**A is unlikely to be provided. Even when a lens that can provide sufficient resolution can be designed, the lens has a problem of low mass producibility because the lens needs to be manufactured with high molding precision. When (OAL/imy)×(LL/imy)×TR×(1/NA) in Conditional Expression (2) is greater than the upper limit, the axial inter-surface spacing from the liquid crystal panel **18** to the reflection surface **40** and the lens diameter of the first lens **34** become excessively large. That is, the amount of protrusion by which the first lens **34** protrudes radially from the optical axis N increases, resulting in an increase in the diameter of the entire projection system. The size of the projector that incorporates the projection system therefore increases.

Example 3 described in JP-A-2020-34690, which is a related-art literature, will now be examined as Comparable Example. The projection system according to Comparable Example includes a first refractive optical system, a reflective optical system, and a second refractive optical system sequentially arranged from the reduction side toward the enlargement side. The first refractive optical system includes a plurality of refractive lenses. The reflective optical system includes a concave mirror and reflects beams from the first refractive optical system toward the side facing the image display device in directions that intersect with the optical axis of the first refractive optical system. The second refractive optical system is formed of a single refractive lens. The refractive lens is an enlargement-side lens located at a position closest to the enlargement side in the projection system. Beams from the concave mirror enter the enlargement-side lens in directions that intersect with the optical axis of the enlargement-side lens. Data on Comparable Example are listed below.

OAL	256 mm
imy	13.2 mm
LL	79.7 mm
TR	0.154
NA	0.25

In Comparable Example, TR=0.154. The projection system according to Comparable Example therefore satisfies Conditional Expression (1). In Comparable Example, however, (OAL/imy)×(LL/imy)×TR×(1/NA)=72. The projection system according to Comparable Example therefore does not satisfy Conditional Expression (2). Therefore, when the throw ratio is fixed in the present example and Comparative Example, the effective radius of the enlargement-side lens of the projection system according to Comparative Example is greater than the effective radius of the first lens of the projection system **3**A according to the present example. That is, the entire projection system according to Comparative Example has a diameter greater than that of the entire projection system **3**A according to the present example.

In the projection system **3**A according to the present example, the reflection surface **40** is provided with a reflection coating layer (reflection layer). In the configuration in which the reflection surface is provided inside the optical element **33**, the accuracy of the shape of the enlargement-side lens surface, at which the reflection surface is provided, depends on the accuracy of the shape of the optical element **33**. That is, to improve the accuracy of the shape of the enlargement-side lens surface, the accuracy of the shape of the reduction-side lens surface also needs to be improved. In contrast, since the reflection surface **40** of the projection system **3**A according to the present example is provided at the outer surface of the optical element **33**, only the accuracy of the shape of the outer surface of the optical element **33** needs to be improved. The accuracy of the shape of the reflection surface **40** in the present example is therefore readily improved as compared with that in the configuration in which the reflection surface is provided inside the optical element **33**.

In the configuration in which the reflection surface is provided inside the optical element **33**, the optical element **33** is formed, and a reflection coating layer is then formed at the enlargement-side lens surface of the optical element **33** to form the reflection surface. In this process, a support film layer needs to be provided between the reflection coating layer and the enlargement-side lens surface. Although the thus provided support film layer causes the reflection coating layer to be unlikely to peel off the enlargement-side lens surface, the interposed support film layer tends to lower the optical performance of the reflection surface, so that the optical performance of the reflection surface tends to vary in the manufacturing process. In contrast, in the projection system **3**A according to the present example, the support film layer is provided on the side opposite from the reflection surface of the reflection coating layer, whereby the optical performance of the reflection surface **40** is unlikely to deteriorate. Stable optical performance of the reflection surface **40** is therefore likely to be achieved during the manufacture of the optical element **33**.

In the projection system **3**A according to the present example, the lens L**17** (second lens) disposed at a position closest to the enlargement side in the first optical system **31** is formed as a lens separate from the first lens **34**. The lens L**17** is disposed between the reflection surface **40** and the first lens **34** in the direction of the optical axis N. That is, the lens L**17** disposed at a position closest to the enlargement side in the first optical system **31** is disposed inside the second optical system **32** in the direction of the optical axis N, so that the distance between the lens L**17** and the reflection surface **40** decreases. The axial inter-surface spacing from the liquid crystal panel **18** to the reflection surface **40** can thus be shortened, whereby the size of the projection system **3**A can be reduced. Furthermore, the intermediate image **30** is formed between the lens L**17** of the first optical system **31** and the reflection surface **40** of the second optical system **32**. When the distance between the lens L**17** and the reflection surface **40** thus decreases, a variety of aberrations contained in the intermediate image **30** are readily corrected on an image height basis.

The first optical system **31** includes the lens L**17** (second lens) and the lens L**16** (third lens), which is disposed adjacent to the lens L**17** and shifted therefrom toward the reduction side. The lenses L**16** and L**17** each have an aspherical shape. In the projection system **3**A according to the present example, focusing that causes the projection distance to be changed from the short distance to the long distance is performed by moving the lenses L**16** and L**17** toward the enlargement side in the direction of the optical axis N. The projection system **3**A, in which the lenses L**16** and L**17**, which correct a variety of aberrations on an image height basis, are moved in the direction of the optical axis N, therefore allows suppression of occurrence of the variety of aberrations during focusing. Furthermore, in a configuration in which a lens having no aspherical shape is moved in the direction of the optical axis N for focusing, an aspherical lens that corrects a variety of aberrations needs to be separately prepared. In contrast, the present example, in which the lenses L**16** and L**17**, which move during focusing, each have an aspherical shape, allows reduction in the size of the entire projection system.

The first optical system **31** further includes the cemented doublets L**23** and L**24** at the enlargement side of the diaphragm **51**. The chromatic aberrations can therefore be corrected well.

FIG. **3** shows lateral aberrations produced by the projection system **3**A set at the standard distance. FIG. **4** shows the spherical aberration, astigmatism, and distortion produced by the projection system **3**A set at the standard distance.

FIG. **5** shows the spherical aberration, astigmatism, and distortion produced by the projection system **3**A set at the short distance. FIG. **6** shows the spherical aberration, astigmatism, and distortion produced by the projection system **3**A set at the long distance. The projection system **3**A according to the present example produces an enlarged image having suppressed aberrations, as shown in FIGS. **3** to **6**.

Example 2

FIG. **7** is a beam diagram showing beams passing through a projection system **3**B according to Example 2. The projection system **3**B according to the present example is formed of a first optical system **31** and a second optical system **32** sequentially arranged from the reduction side toward the enlargement side, as shown in FIG. **7**. The second optical system **32** is disposed on an optical axis N of the first optical system **31**.

The first optical system **31** is a refractive optical system. The first optical system **31** is formed of sixteen lenses L**1** to L**16**. The lenses L**1** to L**16** are arranged in the presented order from the reduction side toward the enlargement side. A diaphragm **51** is disposed between the lens L**7** and the lens L**8**.

The lens L**6** has aspherical shapes at opposite sides. The lens L**9** has aspherical shapes at opposite sides. The lens L**15** (third lens) has aspherical shapes at opposite sides. The lens L**16** (second lens) has aspherical shapes at opposite sides. The lens L**2** and the lens L**3** are bonded to each other into a cemented doublet L**21**. The lens L**4** and the lens L**5** are bonded to each other into a cemented doublet L**22**. The lens L**10** and the lens L**11** are bonded to each other into a cemented doublet L**23**. The lens L**13** and the lens L**14** are bonded to each other into a cemented doublet L**24**.

The second optical system **32** includes an optical element **33** and a first lens **34**. The optical element **33** and the first lens **34** are arranged in the presented order from the reduction side toward the enlargement side. The optical element **33** has a reflection surface **40**, which faces the reduction side. The reflection surface **40** has a concave shape recessed in the second direction Z**2**. The reflection surface **40** has an aspherical shape. The reflection surface **40** is located at the lower side Y**2** of the optical axis N, as shown in FIG. **7**. The reflection surface **40** is formed by providing the outer surface, in the first direction Z**1**, of the optical element **33** with a reflection coating layer (reflection layer). The reflection surface **40** reflects light at the surface, facing in the direction Z**1**, of the optical element **33**.

The first lens **34** is located at a position shifted from the optical element **33** in the first direction Z**1** and disposed at the upper side Y**1** of the optical axis N. The first lens **34** has negative power. The first lens **34** has a convex enlargement-side surface and a concave reduction-side surface. The first lens **34** has aspherical shapes at opposite sides.

The liquid crystal panel **18** of the image formation unit **2** is disposed in the reduction-side conjugate plane of the projection system **3**B. The screen S is disposed in the enlargement-side conjugate plane of the projection system **3**B.

The liquid crystal panel **18** forms a projection image in an image formation plane perpendicular to the optical axis N of the first optical system **31**. The liquid crystal panel **18** is disposed in a position offset from the optical axis N of the first optical system **31** toward the upper side Y**1**. The projection image is therefore formed in a position offset from the optical axis N toward the upper side Y**1**.

The beams from the liquid crystal panel **18** pass through the first optical system **31** and the second optical system **32** in the presented order. Between the first optical system **31** and the second optical system **32**, the beams pass through the lower side Y**2** of the optical axis N. The beams are therefore directed through the second optical system **32** toward the reflection surface **40**. The beams having reached the reflection surface **40** are deflected back in the first direction Z**1** towards the upper side Y**1**. The beams deflected back by the reflection surface **40** cross the optical axis N toward the upper side Y**1** and travel toward the first lens **34**. The beams passing through the first lens **34** are widened by the first lens **34** and reach the screen S.

The lens L**16** of the first optical system **31** is disposed between the reflection surface **40** and the first lens **34** in the direction of the optical axis N. An intermediate image **30** is formed between the lens L**16** and the reflection surface **40**.

In the projection system **3**B, the portion at the reduction side of the first optical system **31** is a telecentric portion.

The projection system **3**B has a changeable projection distance. When the projection distance is changed, seven lenses of the first optical system **31**, the lenses L**10** to L**16**, are moved along the optical axis N for focusing. In the focusing, the lenses L**12**, L**13**, and L**14** are moved as a unit.

Data on the projection system **3**B are listed below,

NA	0.2778
imy	11.8 mm
scy	1462 mm
PD	288.6 mm
M	124
TR	0.197
OAL	189 mm
LL	36.5 mm

where NA represents the numerical aperture of the liquid crystal panel **18**, imy represents a first distance from the optical axis N to the largest image height at the liquid crystal panel **18**, scy represents a second distance from the optical axis N to the largest image height of the enlarged image projected on the screen S, PD represents a projection distance that is the distance from the first lens **34** to the screen S, M represents a projection magnification that is the quotient of division of the second distance by the first distance, TR represents a throw ratio that is the quotient of division of the projection distance by the second distance, OAL represents the axial inter-surface spacing from the liquid crystal panel **18** to the reflection surface **40**, and LL represents the largest radius of the first lens **34**.

Data on the lenses of the projection system **3**B are listed below. The surfaces of the lenses are numbered sequentially from the reduction side toward the enlargement side. Reference characters are given to the liquid crystal panel, the dichroic prism, the lenses, the optical element, the first lens, and the screen. Data labeled with a surface number that does not correspond to any of the liquid crystal panel, the dichroic prism, the lenses, the optical element, the first lens, and the screen is dummy data. Reference character R represents the radius of curvature. Reference character D represents the axial inter-surface spacing. Reference character C represents the aperture radius, and twice the aperture radius is the diameter of the lens surface. Reference characters R, D, and C are each expressed in millimeters.

Reference character	Surface number	Shape	R	D	Glass material	Refraction/ Reflection	C
18	0	Spherical	Infinity	9.5000		Refraction	0.0000
19	1	Spherical	Infinity	25.9100	SBSL7_OHARA	Refraction	12.8706
	2	Spherical	Infinity	0.0000		Refraction	14.7871
L1	3	Spherical	23.3077	7.9094	SFPL51_OHARA	Refraction	15.4467
	4	Spherical	−127.0559	0.1000		Refraction	15.2079
L2	5	Spherical	21.0056	5.4283	SFPL51-OHARA	Refraction	13.0000
L3	6	Spherical	44.2633	1.2142	STIH6_OHARA	Refraction	11.8639
	7	Spherical	26.7444	0.1000		Refraction	11.0399
L4	8	Spherical	17.2046	7.4036	SFPL51_OHARA	Refraction	10.5313
L5	9	Spherical	−23.7897	0.7500	TAFD25_HOYA	Refraction	9.6431
	10	Spherical	20.7471	0.2000		Refraction	8.5696
L6	11	Aspherical	14.2713	4.2460	LBAL35_OHARA	Refraction	8.6504
	12	Aspherical	32.7925	1.0000		Refraction	7.7784
L7	13	Spherical	16.4731	2.7568	SFSL5_OHARA	Refraction	7.6827
	14	Spherical	23.3002	2.7433		Refraction	7.2371
51	15	Spherical	Infinity	0.1000		Refraction	6.9632
L8	16	Spherical	55.5146	3.5141	STIH53_OHARA	Refraction	7.2124
	17	Spherical	−20.4634	0.1000		Refraction	7.4622
L9	18	Aspherical	−18.8417	4.8490	LLAM60_OHARA	Refraction	7.4571
	19	Aspherical	138.4003	Variable spacing 1		Refraction	8.3506
L10	20	Spherical	77.9934	3.2176	STIL25_OHARA	Refraction	13.5492
L11	21	Spherical	−238.3916	1.0000	STIH6_OHARA	Refraction	13.7984
	22	Spherical	422.0121	Variable spacing 2		Refraction	14.0280
L12	23	Spherical	35.5651	10.2093	STIM22_OHARA	Refraction	15.5000
	24	Spherical	−34.9646	0.7658		Refraction	15.5950
L13	25	Spherical	−38.6490	5.7518	STIL25_OHARA	Refraction	15.1870
L14	26	Spherical	−19.2778	1.0000	STIH6_OHARA	Refraction	15.1633
	27	Spherical	431.4628	Variable spacing 3		Refraction	16.3418

-continued

Reference character	Surface number	Shape	R	D	Glass material	Refraction/ Reflection	C
L15	28	Aspherical	−22.5593	3.0000	'Z-E48R'	Refraction	16.5057
	29	Aspherical	111.9646	Variable spacing 4		Refraction	18.7911
L16	30	Aspherical	245.8978	8.0000	'Z-E48R'	Refraction	24.6934
	31	Aspherical	58.0779	Variable spacing 5		Refraction	
40	32	Aspherical	−28.3812	−46.6942		Reflection	39.1428
34	33	Aspherical	33.3357	−7.0000	'Z-E48R'	Refraction	27.0468
	34	Aspherical	30.6616	Variable spacing 6		Refraction	36.2389
S	35	Spherical	Infinity	0.0000		Refraction	2001.4151

The projection system **3**B according to the present example has a changeable projection distance selected from a standard distance, a short distance shorter than the standard distance, and a long distance longer than the standard distance. When the projection distance is changed, seven lenses of the first optical system **31**, the lenses L**10** to L**16**, are moved along the optical axis N for focusing. When the focusing is performed so as to change the projection distance from the short distance to the long distance, the lenses L**10** and L**11** move along the optical axis N toward the reduction side. In the same focusing operation, the lenses L**12**, L**13**, and L**14** move along the optical axis N toward the enlargement side. In the same focusing operation, the lens L**15** moves along the optical axis N toward the enlargement side. In the same focusing operation, the lens L**16** moves along the optical axis N toward the enlargement side.

The table below shows the variable spacings 1, 2, 3, 4, 5, and 6 at the projection distances where the focusing is performed. The variable spacing 1 is the axial inter-surface spacing between the lens L**9** and the lens L**10**. The variable spacing 2 is the axial inter-surface spacing between the lens L**11** and the lens L**12**. The variable spacing 3 is the axial inter-surface spacing between the lens L**14** and the lens L**15**. The variable spacing 4 is the axial inter-surface spacing between the lens L**15** and the lens L**16**. The variable spacing 5 is the axial inter-surface spacing between the lens L**16** and the reflection surface **40**. The variable spacing 6 is the projection distance.

	Standard distance	Short distance	Long distance
Variable spacing 1	12.1167	13.2136	11.8202
Variable spacing 2	1.8056	0.1000	2.7521
Variable spacing 3	5.1195	5.3664	4.9687
Variable spacing 4	20.8516	21.2210	20.3846
Variable spacing 5	38.8253	38.8412	38.8165
Variable spacing 6	−287.0000	−224.0000	−403.0000

The aspherical coefficients are listed below.

Surface number	S11	S12	S18	S19
Radius of curvature (R)	14.2713	32.7925	−18.8417	138.4003
Conic constant (K)	8.61373E−01	1.26960E+01	−1	−90
Fourth-order	−1.78411E−05	1.08502E−04	6.36736E−05	9.09981E−05
Sixth-order	1.92270E−08	3.62039E−07	−9.07712E−07	−7.42228E−07
Eighth-order	8.84500E−11	−2.22710E−09	3.81595E−09	3.02545E−09
Tenth-order		4.90294E−11		

Surface number	S28	S29	S30	S31
Radius of curvature (R)	−22.5593	111.9646	245.8978	58.0779
Conic constant (K)	0	0	90	0.00000E+00
Fourth-order	1.20307E−04	1.83761E−05	−5.27611E−05	−8.10795E−05
Sixth-order	−6.19796E−07	−2.79021E−07	1.51125E−07	2.09269E−07
Eighth-order	1.66495E−09	6.93645E−10	−3.07259E−10	−3.89250E−10
Tenth-order	−1.73986E−12	−6.65573E−13	4.39273E−13	4.46405E−13
Twelfth-order			−2.61547E−16	−1.98979E−16
Fourteenth-order				6.56482E−22

Surface number	S32	S33	S34
Radius of curvature (R)	−28.3812	33.3357	30.66161366
Conic constant (K)	−1.00000E+00	−5.39402E−01	−0.60807081
Fourth-order	3.57234E−06	1.50335E−04	4.29399E−05
Sixth-order	−6.79154E−09	−9.61493E−07	−2.02768E−07
Eighth-order	9.60704E−12	3.12494E−09	3.95931E−10

-continued

Tenth-order	−7.92734E−15	−5.89295E−12	−4.31475E−13
Twelfth-order	3.36716E−18	6.57780E−15	2.74639E−16
Fourteenth-order	−6.07598E−22	−4.02279E−18	−9.56386E−20
Sixteenth-order		1.04005E−21	1.41923E−23

The projection system **3**B according to the present example satisfies all Conditional Expressions (1) and (2) below,

$$TR \leq 0.3 \tag{1}$$

$$35 \leq (OAL/imy) \times (LL/imy) \times TR \times (1/NA) \leq 60 \tag{2}$$

where OAL represents the axial inter-surface spacing from the liquid crystal panel **18** to the reflection surface **40**, imy represents the first distance from the optical axis N to the largest image height at the liquid crystal panel **18**, LL represents the largest radius of the first lens **34**, TR represents the throw ratio, which is the quotient of division of the projection distance by the second distance from the optical axis N to the largest image height of the enlarged image projected on the screen S, and NA represents the numerical aperture of the liquid crystal panel **18**.

Furthermore, it is more preferable that the projection system **3**B satisfies all Conditional Expressions (1) and (2') below.

$$TR \leq 0.3 \tag{1}$$

$$35 \leq (OAL/imy) \times (LL/imy) \times TR \times (1/NA) \leq 53 \tag{2'}$$

In the present example, the values described above are listed below.

OAL	189 mm
imy	11.8 mm
LL	36.5 mm
TR	0.197
NA	0.2778

TR=0.197 is provided from the table shown above, so that Conditional Expression (1) is satisfied. (OAL/imy)×(LL/imy)×TR×(1/NA)=35 is satisfied, so that Conditional Expression (2) is satisfied.

Effects and Advantages

In the projection system **3**B according to the present example, the reflection surface **40** is provided with a reflection coating layer (reflection layer). The projection system **3**B according to the present example can therefore provide the same effects and advantages as those provided by Example 1.

In the projection system **3**B according to the present example, the lens L**16** (second lens) disposed at a position closest to the enlargement side in the first optical system **31** is formed as a lens separate from the first lens **34**. The lens L**16** is disposed between the reflection surface **40** and the first lens **34** in the direction of the optical axis N. That is, the lens L**16** disposed at a position closest to the enlargement side in the first optical system **31** is disposed inside the second optical system **32** in the direction of the optical axis N, so that the distance between the lens L**16** and the reflection surface **40** decreases. The projection system **3**B according to the present example can therefore provide the same effects and advantages as those provided by Example 1.

The first optical system **31** includes the lens L**16** (second lens) and the lens L**15** (third lens), which is disposed adjacent to the lens L**16** and shifted therefrom toward the reduction side. The lenses L**15** and L**16** each have an aspherical shape. In the projection system **3**B according to the present example, focusing that causes the projection distance to be changed from the short distance to the long distance is performed by moving the lenses L**15** and L**16** toward the enlargement side in the direction of the optical axis N. The projection system **3**B, in which the lenses L**15** and L**16**, which correct a variety of aberrations on an image height basis, are moved in the direction of the optical axis N, therefore allows suppression of occurrence of the variety of aberrations during focusing. Furthermore, in a configuration in which a lens having no aspherical shape is moved in the direction of the optical axis N for focusing, an aspherical lens that corrects a variety of aberrations needs to be separately prepared. In contrast, the present example, in which the lenses L**15** and L**16**, which move during focusing, each have an aspherical shape, allows reduction in the size of the entire projection system.

The first optical system **31** further includes the cemented doublets L**23** and L**24** at the enlargement side of the diaphragm **51**. The chromatic aberrations can therefore be corrected well.

The projection system **3**B according to the present example, which satisfies Conditional Expressions (1) and (2), can provide the same effects and advantages as those provided by the projection system **3**A according to Example 1. FIG. **8** shows the lateral aberrations produced by the projection system **3**B set at the standard distance. FIG. **9** shows the spherical aberration, astigmatism, and distortion produced by the projection system **3**B set at the standard distance. FIG. **10** shows the spherical aberration, astigmatism, and distortion produced by the projection system **3**B set at the short distance. FIG. **11** shows the spherical aberration, astigmatism, and distortion produced by the projection system **3**B set at the long distance. The projection system **3**B according to the present example produces an enlarged image having suppressed aberrations, as shown in FIGS. **8** to **11**.

Example 3

FIG. **12** is a beam diagram showing beams passing through a projection system **3**C according to Example 3. The projection system **3**C according to the present example is formed of a first optical system **31** and a second optical system **32** sequentially arranged from the reduction side toward the enlargement side, as shown in FIG. **12**. The second optical system **32** is disposed on an optical axis N of the first optical system **31**.

The first optical system **31** is a refractive optical system. The first optical system **31** is formed of thirteen lenses L**1** to L**13**. The lenses L**1** to L**13** are arranged in the presented order from the reduction side toward the enlargement side. A diaphragm **51** is disposed between the lens L**6** and the lens L**7**.

The lens L**5** has aspherical shapes at opposite sides. The lens L**8** has aspherical shapes at opposite sides. The lens L**12** (third lens) has aspherical shapes at opposite sides. The lens L**13** (second lens) has aspherical shapes at opposite sides. The lens L**3** and the lens L**4** are bonded to each other into a cemented doublet L**21**. The lens L**10** and the lens L**11** are bonded to each other into a cemented doublet L**22**.

The second optical system **32** includes an optical element **33** and a first lens **34**. The optical element **33** and the first lens **34** are arranged in the presented order from the reduction side toward the enlargement side. The optical element **33** has a reflection surface **40**, which faces the reduction side. The reflection surface **40** has a concave shape recessed in the second direction Z**2**. The reflection surface **40** has an aspherical shape. The reflection surface **40** is located at the lower side Y**2** of the optical axis N, as shown in FIG. **12**. The reflection surface **40** is formed by providing the outer surface, in the first direction Z**1**, of the optical element **33** with a reflection coating layer (reflection layer). The reflection surface **40** reflects light at the surface, facing in the direction Z**1**, of the optical element **33**.

The first lens **34** is located at a position shifted from the optical element **33** in the first direction Z**1** and disposed at the upper side Y**1** of the optical axis N. The first lens **34** has negative power. The first lens **34** has a convex enlargement-side surface and a concave reduction-side surface. The first lens **34** has aspherical shapes at opposite sides.

The liquid crystal panel **18** of the image formation unit **2** is disposed in the reduction-side conjugate plane of the projection system **3**C. The screen S is disposed in the enlargement-side conjugate plane of the projection system **3**C.

The liquid crystal panel **18** forms a projection image in an image formation plane perpendicular to the optical axis N of the first optical system **31**. The liquid crystal panel **18** is disposed in a position offset from the optical axis N of the first optical system **31** toward the upper side Y**1**. The projection image is therefore formed in a position offset from the optical axis N toward the upper side Y**1**.

The beams from the liquid crystal panel **18** pass through the first optical system **31** and the second optical system **32** in the presented order. Between the first optical system **31** and the second optical system **32**, the beams pass through the lower side Y**2** of the optical axis N. The beams are therefore directed through the second optical system **32** toward the reflection surface **40**. The beams having reached the reflection surface **40** are deflected back in the first direction Z**1** towards the upper side Y**1**. The beams deflected back by the reflection surface **40** cross the optical axis N toward the upper side Y**1** and travel toward the first lens **34**. The beams passing through the first lens **34** are widened by the first lens **34** and reach the screen S.

The lens L**13** of the first optical system **31** is disposed between the reflection surface **40** and the first lens **34** in the direction of the optical axis N. An intermediate image **30** is formed between the lens L**13** and the reflection surface **40**.

In the projection system **3**C, the portion at the reduction side of the first optical system **31** is a telecentric portion.

The projection system **3**C has a changeable projection distance. When the projection distance is changed, four lenses of the first optical system **31**, the lenses L**9** to L**12**, are moved along the optical axis N for focusing. In the focusing, the lenses L**10** and L**11** are moved as a unit.

Data on the projection system **3**C are listed below,

NA	0.2084
imy	11.7 mm
scy	1462 mm
PD	378.0 mm
M	125
TR	0.259
OAL	172 mm
LL	37.8 mm

where NA represents the numerical aperture of the liquid crystal panel **18**, imy represents a first distance from the optical axis N to the largest image height at the liquid crystal panel **18**, scy represents a second distance from the optical axis N to the largest image height of the enlarged image projected on the screen S, PD represents a projection distance that is the distance from the first lens **34** to the screen S, M represents a projection magnification that is the quotient of division of the second distance by the first distance, TR represents a throw ratio that is the quotient of division of the projection distance by the second distance, OAL represents the axial inter-surface spacing from the liquid crystal panel **18** to the reflection surface **40**, and LL represents the largest radius of the first lens **34**.

Data on the lenses of the projection system **3**C are listed below. The surfaces of the lenses are numbered sequentially from the reduction side toward the enlargement side. Reference characters are given to the liquid crystal panel, the dichroic prism, the lenses, the optical element, the first lens, and the screen. Data labeled with a surface number that does not correspond to any of the liquid crystal panel, the dichroic prism, the lenses, the optical element, the first lens, and the screen is dummy data. Reference character R represents the radius of curvature. Reference character D represents the axial inter-surface spacing. Reference character C represents the aperture radius, and twice the aperture radius is the diameter of the lens surface. Reference characters R, D, and C are each expressed in millimeters.

Reference character	Surface number	Shape	R	D	Glass material	Refraction/ Reflection	C
18	0	Spherical	Infinity	5.1000		Refraction	0.0000
19	1	Spherical	Infinity	23.9250	SBSL7_OHARA	Refraction	12.0558
	2	Spherical	Infinity	0.0000		Refraction	13.1538
L1	3	Spherical	25.3616	5.2165	SFPL51_OHARA	Refraction	13.4219
	4	Spherical	−160.4483	0.1000		Refraction	13.2933
L2	5	Spherical	20.5113	4.3359	SFSL5_OHARA	Refraction	12.0000
	6	Spherical	132.0437	0.0000		Refraction	11.5962
	7	Spherical	Infinity	0.1000		Refraction	11.9120
L3	8	Spherical	20.3053	7.3344	SBSL7_OHARA	Refraction	10.1626
L4	9	Spherical	−23.1488	0.6000	TAFD25_HOYA	Refraction	8.3549
	10	Spherical	13.5441	0.2000		Refraction	7.2414

-continued

Reference character	Surface number	Shape	R	D	Glass material	Refraction/ Reflection	C
L5	11	Aspherical	10.7115	2.4947	LBAL35_OHARA	Refraction	7.2839
	12	Aspherical	21.0609	8.3786		Refraction	6.9453
L6	13	Spherical	34.9176	3.6553	SFSL5_OHARA	Refraction	6.3040
	14	Spherical	−16.1488	0.0364		Refraction	6.0810
51	15	Spherical	Infinity	0.1000		Refraction	5.5409
L7	16	Spherical	−371.7544	2.2081	STIH53_OHARA	Refraction	5.5362
	17	Spherical	−27.0329	0.1512		Refraction	5.4751
L8	18	Aspherical	−23.1991	2.3150	LLAM60_OHARA	Refraction	5.4355
	19	Aspherical	89.7692	Variable spacing 1		Refraction	5.5698
L9	20	Spherical	−64.9866	5.7941	STIM22_OHARA	Refraction	15.7747
	21	Spherical	−27.0076	Variable spacing 2		Refraction	16.5319
L10	22	Spherical	43.6185	10.6351	STIL25_OHARA	Refraction	17.8154
L11	23	Spherical	−30.6037	1.0000	STIH6_OHARA	Refraction	17.7153
	24	Spherical	66.9444	Variable spacing 3		Refraction	17.8736
L12	25	Aspherical	−49.3948	6.5927	‘Z-E48R’	Refraction	19.0969
	26	Aspherical	33.1964	Variable spacing 4		Refraction	18.5842
L13	27	Aspherical	198.8399	5.0000	‘Z-E48R’	Refraction	18.7753
	28	Aspherical	41.3371	35.9841		Refraction	19.8477
40	29	Aspherical	−28.5883	−41.4150		Reflection	29.2645
34	30	Aspherical	38.8043	−5.0000	‘Z-E48R’	Refraction	27.5293
	31	Aspherical	39.6045	Variable spacing 5		Refraction	36.7982
S	32	Spherical	Infinity	0.0000		Refraction	1983.0341

The projection system **3**C according to the present example has a changeable projection distance selected from a standard distance, a short distance shorter than the standard distance, and a long distance longer than the standard distance. When the projection distance is changed, four lenses of the first optical system **31**, the lenses L**9** to L**12**, are moved along the optical axis N for focusing. When the focusing is performed so as to change the projection distance from the short distance to the long distance, the lens L**9** moves along the optical axis N toward the enlargement side. In the same focusing operation, the lenses L**10** and L**11** move along the optical axis N toward the enlargement side. In the same focusing operation, the lens L**12** moves along the optical axis N toward the enlargement side. In the projection system **3**C according to the present example, the lens L**13** is fixed.

The table below shows the variable spacings 1, 2, 3, 4, and 5 at the projection distances where the focusing is performed. The variable spacing 1 is the axial inter-surface spacing between the lens L**8** and the lens L**9**. The variable spacing 2 is the axial inter-surface spacing between the lens L**9** and the lens L**10**. The variable spacing 3 is the axial inter-surface spacing between the lens L**11** and the lens L**12**. The variable spacing 4 is the axial inter-surface spacing between the lens L**12** and the lens L**13**. The variable spacing 5 is the projection distance.

	Standard distance	Short distance	Long distance
Variable spacing 1	27.7058	27.3443	27.9475
Variable spacing 2	0.2085	0.0000	0.4035
Variable spacing 3	5.0259	5.2821	4.8227
Variable spacing 4	8.2762	8.5655	8.0181
Variable spacing 5	−379.0000	−297.0000	−524.0000

The aspherical coefficients are listed below.

Surfaces number	S11	S12	S18	S19
Radius of curvature (R)	10.7115	21.0609	−23.1991	89.7692
Conic constant (K)	2.69624E−01	4.78408E+00	−1	−90
Fourth-order	−7.67687E−05	5.05163E−05	1.47441E−04	2.11126E−04
Sixth-order	−5.77986E−07	−3.02482E−07	−2.37490E−06	−2.24828E−06
Eighth-order	−4.01080E−09	−4.23268E−09	1.45245E−08	1.46576E−08
Tenth-order		5.29887E−11		

-continued

Surface number	S25	S26	S27	S28
Radius of curvature (R)	−49.3948	33.1964	198.8399	41.3371
Conic constant (K)	0	0	90	0.00000E+00
Fourth-order	6.80021E−05	−4.70087E−05	−6.65376E−05	−1.21300E−04
Sixth-order	−9.65755E−08	1.44566E−07	−7.25627E−09	4.65229E−07
Eighth-order	7.76122E−11	−3.72543E−10	6.19846E−10	−1.09606E−09
Tenth-order	2.18093E−14	4.76925E−13	−1.08500E−12	1.26985E−12
Twelfth-order			3.76822E−16	2.28175E−16
Fourteenth-order				−1.41383E−18

Surface number	S29	S30	S31
Radius of curvature (R)	−28.5883	38.8043	39.60445722
Conic constant (K)	−1.00000E+00	4.72790E−01	−0.611371609
Fourth-order	3.91548E−06	−1.70359E−05	−2.40495E−05
Sixth-order	−1.20322E−08	8.73558E−08	5.12452E−08
Eighth-order	2.02386E−11	−2.61450E−10	−6.11798E−11
Tenth-order	−2.19615E−14	5.37788E−13	4.72800E−14
Twelfth-order	1.28670E−17	−6.50975E−16	−2.21909E−17
Fourteenth-orde	−3.20304E−21	4.05800E−19	5.45183E−21
Sixteenth-order		−9.14385E−23	−3.94213E−25

The projection system **3**C according to the present example satisfies all Conditional Expressions (1) and (2) below,

$$TR \leq 0.3 \tag{1}$$

$$35 \leq (OAL/imy) \times (LL/imy) \times TR \times (1/NA) \leq 60 \tag{2}$$

where OAL represents the axial inter-surface spacing from the liquid crystal panel **18** to the reflection surface **40**, imy represents the first distance from the optical axis N to the largest image height at the liquid crystal panel **18**, LL represents the largest radius of the first lens **34**, TR represents the throw ratio, which is the quotient of division of the projection distance by the second distance from the optical axis N to the largest image height of the enlarged image projected on the screen S, and NA represents the numerical aperture of the liquid crystal panel **18**.

In the present example, the values described above are listed below.

OAL	172 mm
imy	11.7 mm
LL	37.8 mm
TR	0.259
NA	0.2084

TR=0.259 is provided from the table shown above, so that Conditional Expression (1) is satisfied. (OAL/imy)×(LL/imy)×TR×(1/NA)=59 is satisfied, so that Conditional Expression (2) is satisfied.

Effects and Advantages

In the projection system **3**C according to the present example, the reflection surface **40** is provided with a reflection coating layer (reflection layer). The projection system **3**C according to the present example can therefore provide the same effects and advantages as those provided by Example 1.

In the projection system **3**C according to the present example, the lens L**13** (second lens) disposed at a position closest to the enlargement side in the first optical system **31** is formed as a lens separate from the first lens **34**. The lens L**13** is disposed between the reflection surface **40** and the first lens **34** in the direction of the optical axis N. That is, the lens L**13** disposed at a position closest to the enlargement side in the first optical system **31** is disposed inside the second optical system **32** in the direction of the optical axis N, so that the distance between the lens L**13** and the reflection surface **40** decreases. The projection system **3**C according to the present example can therefore provide the same effects and advantages as those provided by Example 1.

In the projection system **3**C according to the present example, the first optical system **31** includes the lens L**13** (second lens) and the lens L**12** (third lens), which is disposed adjacent to the lens L**13** and shifted therefrom toward the reduction side. That is, the lens L**13** disposed at a position closest to the enlargement side in the first optical system **31** is disposed inside the second optical system **32** in the direction of the optical axis N, so that the distance between the lens L**13** and the reflection surface **40** decreases. In the projection system **3**C according to the present example, focusing that causes the projection distance to be changed from the short distance to the long distance is performed by moving the lens L**12** toward the enlargement side in the direction of the optical axis N. The projection system **3**C, in which the lens L**12**, which corrects a variety of aberrations on an image height basis, is moved in the direction of the optical axis N, therefore allows suppression of occurrence of the variety of aberrations during focusing. Furthermore, in a configuration in which a lens having no aspherical shape is moved in the direction of the optical axis N for focusing, an aspherical lens that corrects a variety of aberrations needs to be separately prepared. In contrast, the present example, in which the lens L**12**, which moves during focusing, has an aspherical shape, allows reduction in the size of the entire projection system.

The lens L**13** is fixed in the direction of the optical axis N. The lens L**13**, which is the second lens, is disposed between the reflection surface **40** and the first lens **34**. Therefore, when the lens L**13** is moved in the direction of the optical axis N for focusing, the mechanism that moves the lens L**13** is complicated, resulting in an increase in manufacturing cost. The manufacturing cost of the projection system **3**C according to the present example can therefore be suppressed as compared with the manufacturing cost of the projection system **3**A according to Example 1, in which the lens L**17**, which is the second lens, is moved in the direction of the optical axis N.

The first optical system **31** further includes the cemented doublet L**22** at the enlargement side of the diaphragm **51**. The chromatic aberrations can therefore be corrected well.

The projection system **3**C according to the present example, which satisfies Conditional Expressions (1) and (2), can provide the same effects and advantages as those provided by the projection system **3**A according to Example 1. FIG. **13** shows the lateral aberrations produced by the projection system **3**C set at the standard distance. FIG. **14** shows the spherical aberration, astigmatism, and distortion produced by the projection system **3**C set at the standard distance. FIG. **15** shows the spherical aberration, astigmatism, and distortion produced by the projection system **3**C set at the short distance. FIG. **16** shows the spherical aberration, astigmatism, and distortion produced by the projection system **3**C set at the long distance. The projection system **3**C according to the present example produces an enlarged image having suppressed aberrations, as shown in FIGS. **13** to **16**.

Example 4

FIG. **17** is a beam diagram showing beams passing through a projection system **3**D according to Example 4. The projection system **3**D according to the present example is formed of a first optical system **31** and a second optical system **32** sequentially arranged from the reduction side toward the enlargement side, as shown in FIG. **17**. The second optical system **32** is disposed on an optical axis N of the first optical system **31**.

The first optical system **31** is a refractive optical system. The first optical system **31** is formed of thirteen lenses L**1** to L**13**. The lenses L**1** to L**13** are arranged in the presented order from the reduction side toward the enlargement side. A diaphragm **51** is disposed between the lens L**6** and the lens L**7**.

The lens L**5** has aspherical shapes at opposite sides. The lens L**8** has aspherical shapes at opposite sides. The lens L**12** (third lens) has aspherical shapes at opposite sides. The lens L**13** (second lens) has aspherical shapes at opposite sides. The lens L**3** and the lens L**4** are bonded to each other into a cemented doublet L**21**. The lens L**10** and the lens L**11** are bonded to each other into a cemented doublet L**22**.

The second optical system **32** includes an optical element **33** and a first lens **34**. The optical element **33** and the first lens **34** are arranged in the presented order from the reduction side toward the enlargement side. The optical element **33** has a first surface **36**, which faces the reduction side, and a second surface **37**, which faces the side opposite from the first surface **36**. The optical element **33** has a reflection coating layer at the second surface **37**. The first surface **36** has a concave shape. The second surface **37** has a convex shape. The optical element **33** has a first transmission surface **41**, a reflection surface **42**, and a second transmission surface **43** sequentially arranged from the reduction side toward the enlargement side. The first transmission surface **41** is provided at the first surface **36**. The first transmission surface **41** has a concave shape. The reflection surface **42** is the reflection coating layer and has a concave shape to which the surface shape of the second surface **37** has been transferred. The reflection surface **42** reflects light within the optical element **33**. The second transmission surface **43** is provided at the first surface **36**. The second transmission surface **43** has a concave shape. The first transmission surface **41**, the reflection surface **42**, and the second transmission surface **43** each have an aspherical shape. The first transmission surface **41**, the reflection surface **42**, and the second transmission surface **43** are located at the lower side Y**2** of the optical axis N, as shown in FIG. **17**.

The first lens **34** is located at a position shifted from the optical element **33** in the first direction Z**1** and disposed at the upper side Y**1** of the optical axis N. The first lens **34** has negative power. The first lens **34** has a convex enlargement-side surface and a concave reduction-side surface. The first lens **34** has aspherical shapes at opposite sides.

The liquid crystal panel **18** of the image formation unit **2** is disposed in the reduction-side conjugate plane of the projection system **3**D. The screen S is disposed in the enlargement-side conjugate plane of the projection system **3**D.

The liquid crystal panel **18** forms a projection image in an image formation plane perpendicular to the optical axis N of the first optical system **31**. The liquid crystal panel **18** is disposed in a position offset from the optical axis N of the first optical system **31** toward the upper side Y**1**. The projection image is therefore formed in a position offset from the optical axis N toward the upper side Y**1**.

The beams from the liquid crystal panel **18** pass through the first optical system **31** and the second optical system **32** in the presented order. Between the first optical system **31** and the second optical system **32**, the beams pass through the lower side Y**2** of the optical axis N. The beams are thus incident on the first transmission surface **41** of the optical element **33**, which forms the second optical system **32**.

The beams having entered the optical element **33** via the first transmission surface **41** travel toward the reflection surface **42**. The beams having reached the reflection surface **42** are deflected back in the first direction Z**1** towards the upper side Y**1**. The beams deflected back by the reflection surface **42** travel toward the second transmission surface **43**. The beams having exited via the second transmission surface **43** cross the optical axis N toward the upper side Y**1** and travel toward the first lens **34**. The beams passing through the first lens **34** are widened by the first lens **34** and reach the screen S.

The lens L**13** of the first optical system **31** is disposed between the reflection surface **42** and the first lens **34** in the direction of the optical axis N. An intermediate image **30** is formed between the lens L**13** and the reflection surface **42**.

In the projection system **3**D, the portion at the reduction side of the first optical system **31** is a telecentric portion.

The projection system **3**D has a changeable projection distance. When the projection distance is changed, four lenses of the first optical system **31**, the lenses L**9** to L**12**, are moved along the optical axis N for focusing.

Data on the projection system **3**D are listed below,

NA	0.25
imy	11.7 mm
scy	1463 mm
PD	376.0 mm
M	125
TR	0.257
OAL	175 mm
LL	40.0 mm

where NA represents the numerical aperture of the liquid crystal panel **18**, imy represents a first distance from the optical axis N to the largest image height at the liquid crystal panel **18**, scy represents a second distance from the optical axis N to the largest image height of the enlarged image projected on the screen S, PD represents a projection distance that is the distance from the first lens **34** to the screen S, M represents a projection magnification that is the quotient of division of the second distance by the first distance, TR represents a throw ratio that is the quotient of division of the projection distance by the second distance, OAL represents the axial inter-surface spacing from the liquid crystal panel **18** to the reflection surface **42**, and LL represents the largest radius of the first lens **34**.

Data on the lenses of the projection system **3**D are listed below. The surfaces of the lenses are numbered sequentially from the reduction side toward the enlargement side. Reference characters are given to the liquid crystal panel, the dichroic prism, the lenses, the optical element, the first lens, and the screen. Data labeled with a surface number that does not correspond to any of the liquid crystal panel, the dichroic prism, the lenses, the optical element, the first lens, and the screen is dummy data. Reference character R represents the radius of curvature. Reference character D represents the axial inter-surface spacing. Reference character C represents the aperture radius, and twice the aperture radius is the diameter of the lens surface. Reference characters R, D, and C are each expressed in millimeters.

The projection system **3**D according to the present example has a changeable projection distance selected from a standard distance, a short distance shorter than the standard distance, and a long distance longer than the standard distance. When the projection distance is changed, four lenses of the first optical system **31**, the lenses L**9** to L**12**, are moved along the optical axis N for focusing. When the focusing is performed so as to change the projection distance from the short distance to the long distance, the lens L**9** moves along the optical axis N toward the reduction side. In the same focusing operation, the lenses L**10** and L**11** move along the optical axis N toward the enlargement side. In the same focusing operation, the lens L**12** moves along the optical axis N toward the enlargement side. In the projection system **3**D according to the present example, the lens L**13** is fixed.

The table below shows the variable spacings 1, 2, 3, 4, and 5 at the projection distances where the focusing is performed. The variable spacing 1 is the axial inter-surface spacing between the lens L**8** and the lens L**9**. The variable spacing 2 is the axial inter-surface spacing between the lens L**9** and the lens L**10**. The variable spacing 3 is the axial inter-surface spacing between the lens L**11** and the lens L**12**. The variable spacing 4 is the axial inter-surface spacing between the lens L**12** and the lens L**13**. The variable spacing 5 is the projection distance.

Reference character	Surface number	Shape	R	D	Glass material	Refraction/ Reflection	C
18	0	Spherical	Infinity	5.1000		Refraction	0.0000
19	1	Spherical	Infinity	23.9250	SBSL7_OHARA	Refraction	12.2082
	2	Spherical	Infinity	0.0000		Refraction	13.7743
L1	3	Spherical	20.7308	7.6535	SFPL51_OHARA	Refraction	14.3491
	4	Spherical	−160.5725	0.1403		Refraction	14.0489
L2	5	Spherical	19.8009	3.4530	SFSL5_OHARA	Refraction	12.0000
	6	Spherical	42.1883	0.0000		Refraction	11.4760
	7	Spherical	Infinity	0.1846		Refraction	12.5433
L3	8	Spherical	26.7793	5.8661	SBSL7_OHARA	Refraction	10.8722
L4	9	Spherical	−23.4816	0.7000	TAFD25_HOYA	Refraction	10.1298
	10	Spherical	23.8348	0.2000		Refraction	8.9729
L5	11	Aspherical	14.1866	3.5452	LBAL35_OHARA	Refraction	8.8797
	12	Aspherical	23.2921	8.5134		Refraction	8.1796
L6	13	Spherical	33.2276	4.4787	SFSL5_OHARA	Refraction	7.8479
	14	Spherical	−16.5041	0.0000		Refraction	7.6677
51	15	Spherical	Infinity	0.1000		Refraction	6.6414
L7	16	Spherical	846.5448	2.2054	STIH53_OHARA	Refraction	6.6229
	17	Spherical	−40.4470	0.2313		Refraction	6.4863
L8	18	Aspherical	−25.5044	3.6042	LLAM60_OHARA	Refraction	6.4792
	19	Aspherical	59.3482	Variable spacing 1		Refraction	6.1272
L9	20	Spherical	−437.3885	6.0000	STIM22_OHARA	Refraction	17.0050
	21	Spherical	−37.6532	Variable spacing 2		Refraction	17.5773
L10	22	Spherical	31.8187	11.5532	STIL25_OHARA	Refraction	19.3821
L11	23	Spherical	−47.9275	1.3832	STIH6_OHARA	Refraction	19.2050
	24	Spherical	49.3856	Variable spacing 3		Refraction	18.3019
L12	25	Aspherical	−177.2317	2.0000	'Z-E48R'	Refraction	18.4483
	26	Aspherical	22.7112	Variable spacing 4		Refraction	17.9293
L13	27	Aspherical	179.6432	5.0000	'Z-E48R'	Refraction	19.9221
	28	Aspherical	27.1594	35.4677		Refraction	19.7715
41	29	Aspherical	−32.3930	3.0000	'Z-E48R'	Refraction	30.0301
42	30	Aspherical	−30.4878	−3.0000	'Z-E48R'	Reflection	30.7470
43	31	Aspherical	−32.3930	−41.1512		Refraction	29.6535
34	32	Aspherical	39.4467	−5.0000	'Z-E48R'	Refraction	35.7473
	33	Aspherical	38.8749	Variable spacing 5		Refraction	39.9951
S	34	Spherical	Infinity	0.0000		Refraction	1984.0193

	Standard distance	Short distance	Long distance
Variable spacing 1	26.8792	27.0101	26.6608
Variable spacing 2	1.4097	0.7506	2.2492
Variable spacing 3	2.1500	2.6303	1.5624
Variable spacing 4	10.4814	10.5279	10.4465
Variable spacing 5	−376.0000	−295.0000	−519.0000

The aspherical coefficients are listed below.

Surface number	S11	S12	S18	S19
Radius of curvature (R)	14.1866	23.2921	25.5044	59.3482
Conic constant (K)	0.52162588	4.313809097	−1	−90
Fourth-order	−2.90353E−05	1.01864E−04	9.38265E−05	1.86743E−04
Sixth-order	−2.33563E−07	−1.24711E−06		
	−3.97479E−08	−1.68413E−06		
Eighth-order	−7.08031E−09	5.72359E−09	9.15280E−09	
	−1.14146E−08			
Tenth-order		2.69207E−11		

Surface number	S25	S26	S27	S28
Radius of curvature (R)	−177.2317	22.7112	179.6432	27.1594
Conic constant (K)	0	0	89.447202	0.000000
Fourth-order	7.96862E−05	−2.58626E−08	−6.20694E−05	−1.51197E−04
Sixth-order	−3.07379E−07	−1.92895E−07	4.44832E−07	9.77046E−07
Eighth-order	6.58543E−10	2.75182E−10	−2.36330E−09	−4.47460E−09
Tenth-order	−4.83689E−13	−1.73130E−15	5.49541E−12	1.16337E−11
Twelfth-order			−4.88375E−15	−1.62518E−14
Fourteenth-order				9.38335E−18

Surface number	S29, S31	S30	S32	S33
Radius of curvature (R)	−32.3930	−30.4878	39.4467	38.8749
Conic constant (K)	−1.026766	−1.000000	−0.081546	−0.618360
Fourth-order	9.61441E−06	5.12974E−06	−1.83922E−05	−1.83960E−05
Sixth-order	−4.77776E−08	−1.81978E−08	8.40433E−08	3.72462E−08
Eighth-order	1.03751E−10	2.96797E−11	−1.91747E−10	−5.13686E−11
Tenth-order	−1.17673E−13	−2.63402E−14	2.17884E−13	5.06109E−14
Twelfth-order	6.81981E−17	1.12362E−17	−5.07382E−17	−3.06209E−17
Fourteenth-order	−1.55320E−20	−1.48819E−21	−7.93912E−20	1.04462E−20
Sixteenth-order			4.02973E−23	−1.62167E−24

The projection system **3**D according to the present example satisfies all Conditional Expressions (1) and (2) below,

$$TR \leq 0.3 \quad (1)$$

$$35 \leq (OAL/imy) \times (LL/imy) \times TR \times (1/NA) \leq 60 \quad (2)$$

where OAL represents the axial inter-surface spacing from the liquid crystal panel **18** to the reflection surface **42**, imy represents the first distance from the optical axis N to the largest image height at the liquid crystal panel **18**, LL represents the largest radius of the first lens **34**, TR represents the throw ratio, which is the quotient of division of the projection distance by the second distance from the optical axis N to the largest image height of the enlarged image projected on the screen S, and NA represents the numerical aperture of the liquid crystal panel **18**.

Furthermore, it is more preferable that the projection system **3**D satisfies all Conditional Expressions (1) and (2') below.

$$TR \leq 0.3 \quad (1)$$

$$35 \leq (OAL/imy) \times (LL/imy) \times TR \times (1/NA) \leq 53 \quad (2')$$

In the present example, the values described above are listed below.

OAL	175 mm
imy	11.7 mm
LL	40.0 mm
TR	0.257
NA	0.25

TR=0.257 is provided from the table shown above, so that Conditional Expression (1) is satisfied. (OAL/imy)×(LL/imy)×TR×(1/NA)=53 is satisfied, so that Conditional Expression (2) is satisfied.

Effects and Advantages

In the projection system **3**D according to the present example, the lens L**13** (second lens) disposed at a position closest to the enlargement side in the first optical system **31** is formed as a lens separate from the first lens **34**. The lens L**13** is disposed between the reflection surface **42** and the first lens **34** in the direction of the optical axis N. That is, the lens L**13** disposed at a position closest to the enlargement side in the first optical system **31** is disposed inside the second optical system **32** in the direction of the optical axis N, so that the distance between the lens L**13** and the reflection surface **42** decreases. The projection system **3**D according to the present example can therefore provide the same effects and advantages as those provided by Example 1.

In the projection system **3**D according to the present example, the first optical system **31** includes the lens L**13** (second lens) and the lens L**12** (third lens), which is disposed adjacent to the lens L13 and shifted therefrom toward the reduction side. The lenses L12 and L13 each have an aspherical shape. In the projection system 3D according to the present example, focusing that causes the projection distance to be changed from the short distance to the long distance is performed by moving the lens L12 toward the enlargement side in the direction of the optical axis N. The projection system 3D, in which the lens L12, which corrects a variety of aberrations on an image height basis, is moved in the direction of the optical axis N, therefore allows suppression of occurrence of the variety of aberrations during focusing. Furthermore, in a configuration in which a lens having no aspherical shape is moved in the direction of the optical axis N for focusing, an aspherical lens that corrects a variety of aberrations needs to be separately prepared. In contrast, the present example, in which the lens L12, which moves during focusing, has an aspherical shape, allows reduction in the size of the entire projection system.

The lens L13 is fixed in the direction of the optical axis N. The lens L13, which is the second lens, is disposed between the reflection surface 42 and the first lens 34. Therefore, when the lens L13 is moved in the direction of the optical axis N for focusing, the mechanism that moves the lens L13 is complicated, resulting in an increase in manufacturing cost. The manufacturing cost of the projection system 3D according to the present example can therefore be suppressed as compared with the manufacturing cost of the projection system 3A according to Example 1, in which the lens L17, which is the second lens, is moved in the direction of the optical axis N.

The first optical system 31 further includes the cemented doublet L22 at the enlargement side of the diaphragm 51. The chromatic aberrations can therefore be corrected well.

The projection system 3D according to the present example, which satisfies Conditional Expressions (1) and (2), can provide the same effects and advantages as those provided by the projection system 3A according to Example 1. FIG. 18 shows the lateral aberrations produced by the projection system 3D set at the standard distance. FIG. 19 shows the spherical aberration, astigmatism, and distortion produced by the projection system 3D set at the standard distance. FIG. 20 shows the spherical aberration, astigmatism, and distortion produced by the projection system 3D set at the short distance. FIG. 21 shows the spherical aberration, astigmatism, and distortion produced by the projection system 3D set at the long distance. The projection system 3D according to the present example produces an enlarged image having suppressed aberrations, as shown in FIGS. 18 to 21.

What is claimed is:

1. A projection system for enlarging a projection image formed by an image formation device disposed in a reduction-side conjugate plane and projecting the enlarged image in an enlargement-side conjugate plane, the projection system comprising:

a first optical system and a second optical system sequentially arranged from the reduction side toward the enlargement side,

wherein the first optical system includes a diaphragm,

the second optical system includes an optical element having a concave reflection surface and a first lens having negative power, the optical element and the first lens sequentially arranged from the reduction side toward the enlargement side,

an intermediate image conjugate with the reduction-side conjugate plane and the enlargement-side conjugate plane is formed between the first optical system and the second optical system,

a portion at the reduction side of the first optical system forms a telecentric portion, and

the projection system satisfies all Conditional Expressions (1) and (2) below,

$$TR \leq 0.3 \tag{1}$$

$$35 \leq (OAL/imy) \times (LL/imy) \times TR \times (1/NA) \leq 60 \tag{2}$$

where OAL represents an axial inter-surface spacing from the image formation device to the reflection surface, imy represents a first distance from an optical axis to a largest image height at the image formation device, LL represents a largest radius of the first lens, TR represents a throw ratio that is a quotient of division of a projection distance by a second distance from the optical axis to a largest image height of the enlarged image, and NA represents a numerical aperture of the image formation device.

2. The projection system according to claim 1, wherein the reflection surface is provided with a reflection layer.

3. The projection system according to claim 1,

wherein a second lens disposed at a position closest to the enlargement side in the first optical system is formed as a lens separate from the first lens, and

the second lens is disposed between the reflection surface and the first lens in a direction of the optical axis.

4. The projection system according to claim 3,

wherein the first optical system includes the second lens and a third lens that is disposed adjacent to the second lens and shifted therefrom toward the reduction side,

the second and third lenses each have an aspherical shape, and

focusing that causes the projection distance to be changed from a short distance to a long distance is performed by moving the third lens toward the enlargement side in the direction of the optical axis.

5. The projection system according to claim 4, wherein the second lens is fixed in the direction of the optical axis.

6. The projection system according to claim 1, wherein the first optical system includes a cemented doublet on the enlargement side of the diaphragm.

7. A projector comprising:

the projection system according to claim 1; and

the image formation device that forms a projection image in the reduction-side conjugate plane of the projection system.

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